Lecture 126

MS1 & Intact Protein Mass

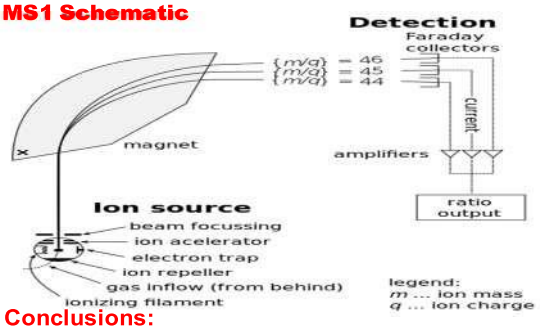
Introduction:

- Once a protein is ionized, it can be deflected by a magnetic field in proportion to its mass
- Proteins can then be measured for their mass in a Mass Spectrometer(MS).

Processing MS1:

- Mass/Charge ratio helps calculate the mass of the protein
- What if a protein carries multiple charges?
- "Mass-select" can help select a specific MS1 for further analysis.

MS1 Schematic



Conclusions:

- MS1 spectra reports the intact protein masses for all proteins in a protein complex
- Incase the sample contains peptides, MS1 gives the intact mass of the peptides

Scoring:

- We can score each protein hit in a way that the better match gets a higher score while the low quality match gets a lower score
- But how to be sure that a peak is from a multiple charged ion?
- So once we have filtered away multiple charged peaks, we retain only those peaks with charge 1
- Next, we filter protein database and obtain candidate hits

Conclusions:

- Experimental mass reported from MS1 is matched with theoretical mass of proteins in the database
- Score is awarded on the basis of the closeness between experimental and theoretical mass

Resolving Multiple Charges

Example Suppose we have a peptide with mass 2000.0 Da, and that the ionization yields peptide ions of charge +1, +2, *and* +3, by the attraction of one, two, or three protons, respectively. The peptide ions will then be detected at

ions with charge +1: $m/z = (2000 + 1)/1 = 2001$
ions with charge +2: $m/z = (2000 + 2)/2 = 1001$
ions with charge +3: $m/z = (2000 + 3)/3 = 666.7$

Scoring Scheme:

$$M_{Score} = \frac{1}{\sqrt{(M_{Exp} - MT)^2}}$$

Example: A Protein Sequence from Database "MQLF"

http://web.expasy.org/compute_pi/ **MW: 537.67**

Lecture 127

Scoring Intact Protein Mass

Background:

- MS1 helps obtain the intact mass of the precursors molecules
- Precursor molecules may be proteins or peptides depending upon the proteomics protocol employed

Introduction:

- Using the masses reported by MS1, protein databases can be filtered for candidate protein matches (or hits)
- Remember to convert the masses of all molecules to 1+mass

Lecture 128

Protein Fragmentation Techniques

Background:

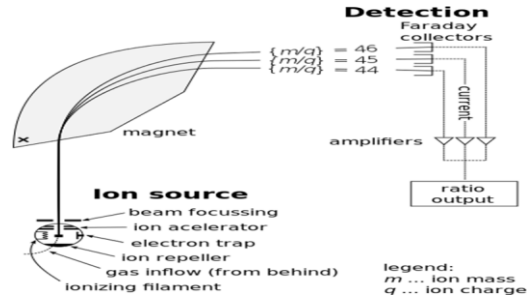
- MS1 helps measure the molecular weight of intact proteins or peptides
- One can compare this mass with the theoretically calculated protein mass from database & score/rank matches.

Introduction:

- What about the case that several proteins obtain the same matching score?
- In this case, an other step may be required to further classify/score proteins in order to obtain the most probable candidate!

- What if we could fragment the ionized intact protein or peptide within the MS chamber?
- This would help measure the fragment masses in the same way as their precursors!

Modifying MS1 to include fragmentation



Lecture 129

Tandem MS

Introduction:

- Intact proteins or peptides can be measured for their intact mass
- This can be followed up by their fragmentation within the MS chamber
- Can we measure the mass of these fragments?

MS2 and MS'n':

- Tandem MS can be extended to the fragments of the intact fragment!
- All you need is the MS instrument capability to
  (i) select fragment's mass range
  (ii) fragment the precursor fragment

Techniques:

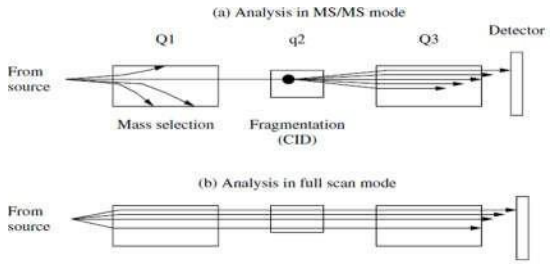
- Electron Capture Dissociation (ECD)
- Electron Transfer Dissociation (ETD)
- Collision Induced Dissociated (CID)

Fragmentation Products:

- Each fragmentation technique results in a specific type of fragment
- ECD gives out 'C' and 'Z' ions
- CID gives out 'B' and 'Y' ions etc..

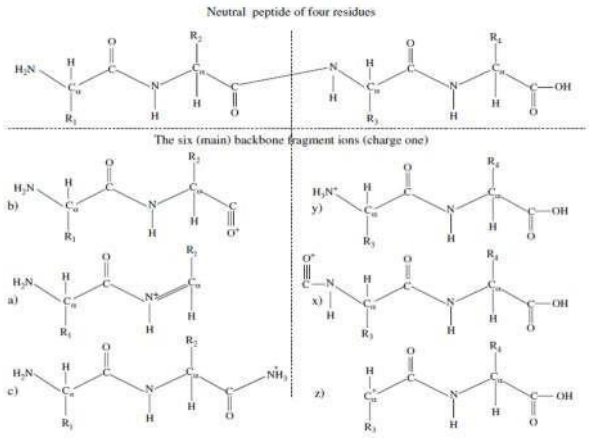
Conclusions: If we can:

1. measure the mass of fragments using MS
2. Calculate the theoretical mass of the fragments
- Then, we can award score on the basis of the similarity of experimental and theoretical mass



Conclusion:

- Tandem MS helps in measurement of mass to the fragments as well!
- This process provides an other step in further scoring and ranking
- Protein identification thus becomes easier.



Lecture 130

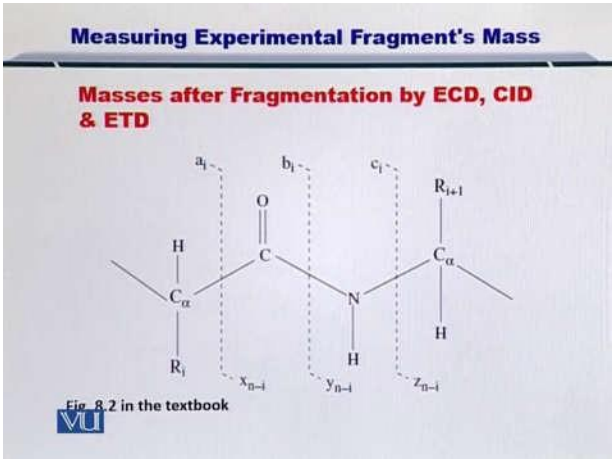
Measuring Experimental Fragment's Mass:

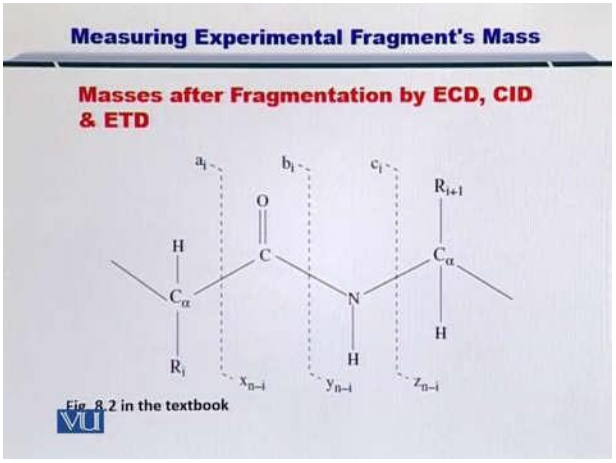
Introduction:

- In MS1, molecular weight of the intact sample molecule can be measured
- The intact sample molecule may be fragmented into 2 afterward
- The two fragments are measured by Tandem MS or MS2

Fragmentation Techniques and Molecular Weight

- Fragmentation techniques include ECD, CID etc.
- Intact molecule's fragmentation splits the intact molecule into two parts
- ECD _>c & z ions



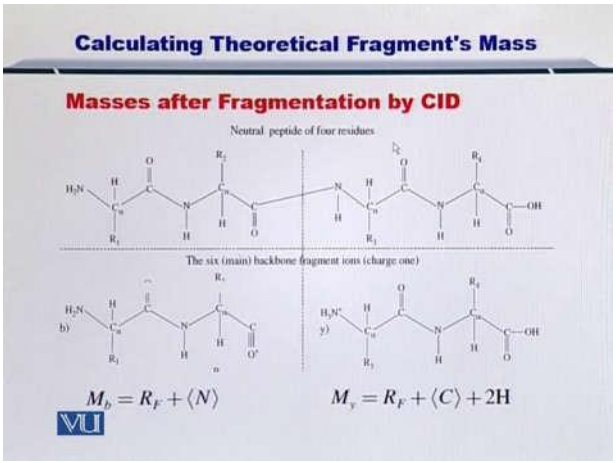


Fragment's Mass:

- Masses of fragments produced by MS2 depend upon the fragmentation technique used
- Because each fragmentation method splits the protein/peptide at a unique location

Conclusion:

- Experimental mass reported from MS2 is matched with theoretical peptides of candidate proteins (from DB)
- Score is awarded on the basis of the closeness between experimental and theoretical masses.



Lecture 131

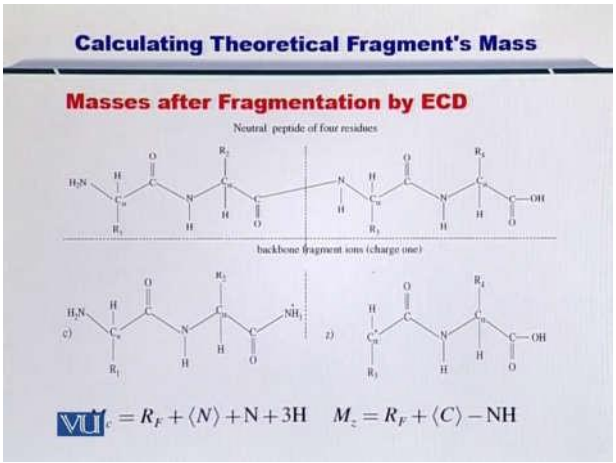
Calculating Theoretical Fragment's Mass:

Introduction:

- In MS1, molecular weight of the intact sample molecule can be measured
- We compare these fragment masses with the theoretical fragments of proteins from sequence database.

Fragmentation Techniques and Molecular Weight

- Fragmentation techniques include ECD, CID etc.
- Depends upon the fragmentation technique, mass of the two theoretical fragments can be calculated.



Conclusion:

- Experimental mass reported from MS2 is matched with theoretical peptides of candidate proteins (from DB)
- Score is awarded on the basis of the closeness between experimental and theoretical masses.

Lecture 132

Protein Sequences

Peptide Sequence Tags

Introduction: **Peptide Sequence Tags (PST) are sequences of peptides produced during MS2**

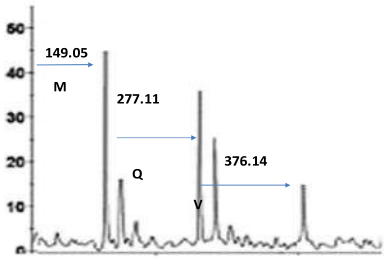
How can we obtain the sequence of a peptide after MS2?

The answer lies in the variation in fragmentation sites

Mass Select: [375.0 , 377.0]

Protein Sequence: **"MQV…"**

Fragments formed may be **M, MQ, QV** etc.



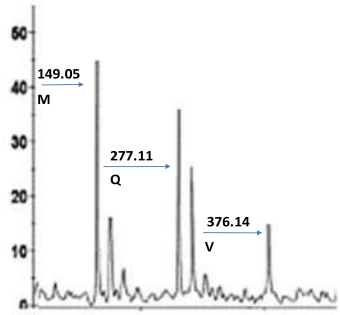
Mass Select: [375.0 , 377.0]

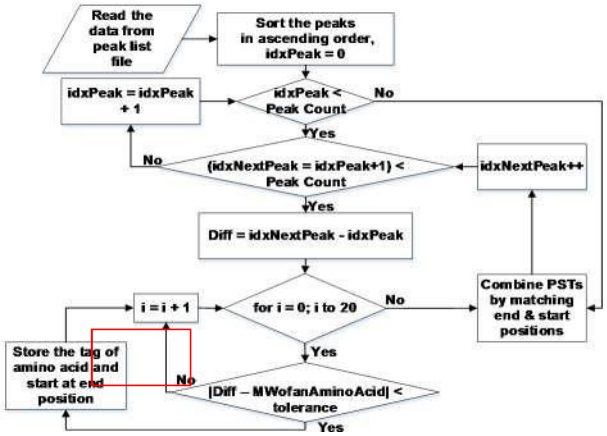
Protein Sequence: **"MQV…"**

Fragments formed may be **M, MQ, QV** etc.

Conclusions:

Precursor proteins or peptides fragmentation leads to formation of multiple ions of the same fragment type However, fragments have variation in their molecular weights due to variation in site of fragmentation





Conclusions:

- Fragmentation at consecutive sites leads to a mass difference equal to that of a single amino acid
- Such consecutive peaks can reveal partial peptide sequence tags.

Lecture 133

Extracting Peptide Sequence Tags

- Peptide Sequence Tags (PST) are sequences of peptides produced during MS2
- PSTs are formed due to sequential cleavage of precursor protein/peptide's backbone

Conclusions: Peptide sequence tags can be extracted from peak list iteratively

- A high quality mass spectrum will produce large number of PSTs
- The bigger the peptide sequence tags, the better!

Lecture 134

Using Peptide Sequence Tags in Protein Search

Introduction:

- Peptide Sequence Tags (PST) are small amino acid sequences of peptides produced during MS2
- PSTs provide clues of the precursor protein/peptides sequence
- How to incorporate them in the search?

Repeating the Process:

- For all the proteins in the database, we find out which PSTs exist in which proteins
- The protein reporting the most PSTs is more probable to be the precursor protein.

- Consider that we extract the following PSTs: M, MQ, QV etc.
- Search protein sequence database (e.g. Uniprot, Swissprot)
  - Sample sequence in protein DB

>>sp|Q6GZ4X|0X1R_FRG3G Putative transcription factor 0X1R OS=Random virus 3 (isolate Goorha) GN=FV3-0X1R PE=4 SV=1

MAFSAEDVLKEYDRRRRMEALLLSLYYPNDRKLLDYKEWSPPRVQVECPKA PVEWNNPPSEKGLIVGHFSGIKYKGEKAQASEVDVNKMCCWVSKFKDAM RRYQGIQTCKIPGQVLSDLDAKIKAYNLTVEGVEGFVRYSRVTKQHVAAFLK ELRHSKQYENVNLIHYILTDKRVDIQHLEKDLVKDFKALVESAHRMRQGHM

QNVKYILYQLLKKHGHGPDGPDILTVKTGSMQLYDDSFRKIYTDLGWKFTPL

Conclusions:
- What if several proteins report an equal No. of PSTs?
- Or what if a protein reports longer PSTs while an other 1 reports smaller yet greater number of PSTs?
- Scoring PSTs?
- We extracted PSTs and searched the entire database for those protein who report these PSTs
- PST search can complement the molecular weight search from MS data.

Lecture 135

Scoring Peptide Sequence Tags

Introduction:
- Peptide Sequence Tags (PST) are small amino acid sequences of peptides produced during MS2
- PSTs provide clues of the precursor protein/peptides sequence
- How to incorporate them in the search?

Scoring Scheme
- If a candidate protein matches 'n' PSTs, then its score can be given by:

$$PST_{Score} = \sum_{i=0}^{n} Length(PST_i)^2$$

- Additionally, if we include RMSE to the scoring system, then it can highlight better PST matches.

RMSE of PSTs • RMSE is the root mean square error

Scoring Scheme
- RMSE for a sequence tag 'i' of length 'n'?

$$RMSE_i = \sum_{i=0}^{n} \sqrt{(M_{Hop} - MAA)\ 2}$$

So, the updated relationship is:

$$PST_{Score} = \sum_{i=0}^{n} \left(\frac{Length(PSTi)}{RMSE_i}\right)^2$$

Conclusions:
- We extracted PSTs and searched the entire database for those protein who report these PSTs
- PST search can complement the molecular weight search from MS data.

Lecture 136

In silico Fragment Comparison

Background:
- MS1 reports the intact mass of molecules (proteins or peptides) in the sample
- Intact mass can be compared with every protein's mass in database to identify the molecule in sample
- Incase multiple candidate proteins are reported, MS2 can be performed
- MS2 helps measure fragment peptide masses
- MS2 data can used to extract peptide sequence tags

Introduction:
- What if the protein identification is still not deterministic?
- Each experimentally reported MS2 fragment is compared with the in silico spectra of proteins from database.

Technique:
- Fragmentation techniques determine product ions e.g. ECD -> c/z and CID -> b/y ions etc.
- With known fragment types, we can compute the MW of all possible protein fragments.

Generating In silico Spectra

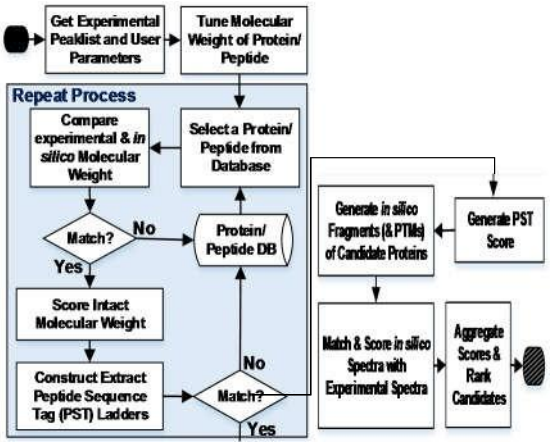
For obtaining all possible theoretical fragments in a protein, we need to compute the MWs of each fragment individually

Consider a random protein sequence from DB:



Conclusions:

- Matching experimental fragments with in silico fragments is the final resort in protein search & identification
- How can we quantify the match b/w in silico & in vitro fragments?



Lecture 137

In silico Fragment Comparison and Scoring

- Experimental MS2 fragments can be compared with the in silico spectra of proteins from database
- How to score these comparisons?

1. Count the matches b/w in silico & in vitro peaks
2. Give an equivalent score to the candidate protein
3. Weigh each of the aforementioned match by the mass error
4. Accumulate the score

Conclusions: With "all possible" fragments in in silico spectrum, & "reported" fragments in experimental spectrum, we can match & rank!

- Scoring scheme should also consider the errors in peak matching

Lecture 138

Protein Sequence Database Search Algorithm

Introduction:

- MS1 and MS2 provide us with a host of data towards enabling us in identifying unknown proteins
- A step by step approach combining MW, PSTs and insilico spectral matching is required

Conclusions:

- Integrating MW, PST and and insilico comparison algorithms in a workflow can help create a composite protein search engine
- A composite scoring system is also required for this search engine

Lecture 139

Integrative Scoring Schemes

Background:

- Three individual scores can be obtained including

1. MW Match Score
2. PST Match Score
3. In vitro & In silico Match Score

Introduction:

- Why is it necessary to compute an overall cumulative score
- What are the options that we have?

Conclusions:

- Composite scoring schemes are needed to combine scores coming in from multiple criteria
- The ability of a scoring scheme to better isolate true positives from false positives is important!

- Simply sum the scores up (a linear function)

$$\text{Score}_{MW} + \text{Score}_{PST} + \text{Score}_{Exp <> TMr} = \text{Score}$$

- Weigh each scoring component up by respective RMSE before summing them up

$$\text{Score} = \frac{\text{Score}_{MW}}{E_{MW}} + \sum_{i=0}^{m} \frac{\text{Score}_{PST}}{RMSE_{PST}} + \sum_{i=0}^{n} \frac{\text{Score}_{Exp <> TMr}}{E_{EXP <> TMr}}$$

- Complex non-linear functions integrate the scoring components in Mascot etc.
  - Highly proprietary for commercial proteomics software

Lecture 140

Large Scale Proteomics

- Peptide mixtures in bottom up proteomics are very complex
- Tryptic peptides may reach up to an order of 300,000–400,000
- In whole proteome samples, protein count may be over 10,000
- Experiments have shown that it is difficult or even impossible to analyze all these peptides in a single analysis, as the mass spectrometer is essentially overwhelmed.
- Over half a million peptides reported in a typical LSP experiment are redundant
- What if we could find one unique peptide for each protein, which could singularly identify the protein?
- However, that would make sequence coverage suffer!
- So the idea is to strike a compromise between sequence coverage and sample coverage!

Technique:

- One way forward would be to transfer peptides to the MS chamber in a step-by-step manner
- However, this imposes a precondition that a peptide is not selected earlier as well (i.e. more than once)

Conclusions:

The number of MS1/2 scans can be limited by carefully selecting the peptide peaks.
Once the intense peptides are identified, next batch of peptides is chosen for MS2.

Step-by-step Technique:

1. The instrument alternates b/w MS & MS/MS modes.
2. 3 most intense peaks are chosen for MS/MS analysis.
3. After the initial MS scan, an MS/MS spectrum from peptide A is obtained by selectively fragmenting this mass only.
4. Next, a spectrum for peptide B is produced, followed by a recording of the MS/MS spectrum for peptide C.
5. After these three fragmentation spectra have been obtained, a new MS scan is started.
6. From this scan, 3 more peptides A B C are selected for fragmentation and the cycle starts over again.

Lecture 141

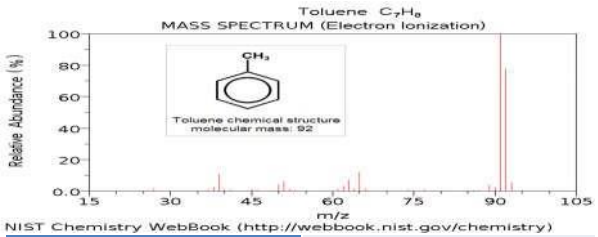
Proteomics Data File Formats

Background:

- Mass spectrometer is used to measure mass/charge ratio of ionized proteins and peptides
- Data output from the MS comprises of m/z ratios and intensities of each molecule that is measured

Conclusions:

- Multiple MS data formats exist
- Proprietary formats exist which come implemented as software with hardware
- Also, open software standards exist for interoperability etc.

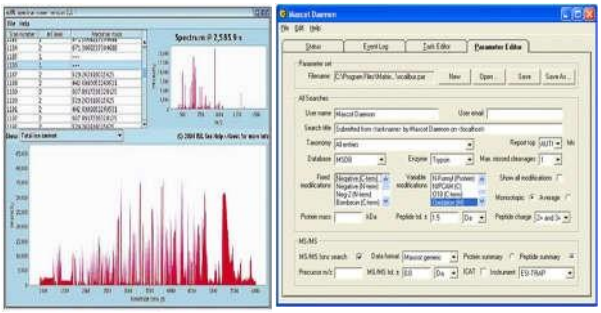


Agilent	.D/.YEP
Bruker	.BAF
ABI/Sciex	.WIFF
	.t2d
Thermo Xcalibur	.RAW
Micromass	
PerkinElmer	
Waters	
	.QSD

Open Formats:

>mzXML (tools.proteomecenter.org/mzXMLViewer.php)

>MGF (proteomicsresource.washington.edu/mascot/help/data_file_help.html)



Lecture 142

RAW File Format

Background:

- Mass spectrometer outputs data with ionic mass/charge ratios & respective ion intensities
- RAW file is a format in which an instrument outputs data in binary form
- Multiple RAW file formats exist

Conclusions:

- Multiple RAW file formats are prevailing in the industry
- Each vendor has its own unique RAW file format
- You can convert proprietary formats into open formats

Vendor	*Instrument*	*Data Type*	*Converter & Set Up*
Thermo	Ion Trap / Orbitrap	raw	MSFileReader
AB SCIEX	Triple TOF	wiff	AB SCIEX Data Converter
	QSTAR / QTRAP	wiff	Analyst + mzWiff
	TOF TOF	wiff	AB Extractor
Agilent	Ion Trap	yep	CompassXtract
	Q-TOF	d	MassHunter
Bruker	QTOF / Ion Trap / TOF TOF	baf / yep / lift	CompassXtract
Shimadzu	AXIMA-CFR	run	Kratos Converter
Varian	FTICR / Ion Trap	sms/xms	VarianMS
Waters	Q-TOF	raw	MassLynx

Free Software Name	Input RAW Data Type	Output Data Types
ProteoWizard	Thermo, Agilent, Bruker, Waters, AB Sciex	mzML, TraML, mzIdentML, mzXML
OpenMS	All vendor formats	mzML, TraML, mzIdentML, mzData, mzQuantML
Trans Proteomic Pipeline	Thermo, ABI Sciex, Agilent, Waters.[14]	mzML, mzXML, pepXML, protXML (Proteowizard)
PEAKS	Thermo, ABI Sciex, Agilent, Waters, Bruker, Shimadzu, Varian	MSF, tandem, mzML, omx, dat, FASTA

Lecture 143

MGF File Format

Background:

- MGF – Mascot Generic Format
- MGF is a simple human-readable format for MS/MS data developed by Matrix Science
- Mascot Search Engine available at http://www.matrixscience.com/search_form_select.html

Conclusions:

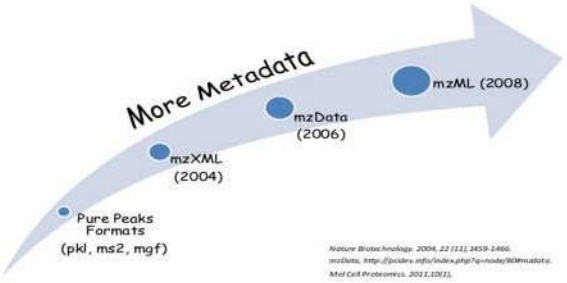
- MGF file format was developed as an open standard for proteomics data
- However, several versions of MGF file formats now exist
- Before using an MGF file, it is useful to check its version

Name	Description	Header	Local	Choices/Range	Notes
ACCESSION	Database entries to be searched	✔		List of double quoted, comma separated values	
CHARGE	Peptide charge	✔	✔	1-	M-H- on PMF form
				Mr	
				1+	MH+ on PMF form
				N- to N+ where N is an integer and combinations	Not PMF
CLE	Enzyme	✔		Trypsin etc., as defined in enzymes file	No default, so must be specified
COM	Search title	✔			Applies to the whole search
CUTOUT	Precursor removal	✔		Pair of comma separated integers	MIS only
COMP	Amino acid composition		✔		
DB	Database	✔		As defined in mascot.dat	
DECOY	Perform decoy search	✔		0 (false)	Default
				1 (true)	
ERRORTOLERANT	Error tolerant	✔		0 (false)	Default
				1 (true)	Not PMF
ETAG	Error tolerant sequence tag		✔		A single query can have multiple ETAGs

FORMAT	MS/MS data file	✔		Mascot generic	Default
				Sequest (.DTA)	
				Finnigan (.ASC)	
				Micromass (.PKL)	
				PerSeptive (.PKS)	
				Sciex API III	
				Bruker (.XML)	
				mzData (.XML)	
				mzML (.mzML)	
FRAMES	NA translation	✔		Comma separated list of frames	Default is 1,2,3,4,5,6
INSTRUMENT	MS/MS ion series	✔	✔	Default	Default
				ESI-QUAD-TOF etc., as defined in fragmentation_rules	
IT_MODS	Variable Mods	✔	✔	As defined in unimod.xml	
ITOL	Fragment ion tol.	✔		Unit dependent	

Solution:

- mzData was developed by HUPO-PSI
- mzXML was developed at the Institute for Systems Biology
- To combine them, a joint venture produced mzML



ITOLU	Units for ITOL	✔		ppm	
				Da	
				mmu	
LOCUS	Hierarchical scan range identifier		✔	string	MIS only
MASS	Mono. or average	✔		Monoisotopic	
				Average	
MODS	Fixed Mods	✔		As defined in unimod.xml	
MULTI_SITE_MODS	Allow two modifications at a single site	✔		0 (false) or 1 (true)	default 0
PEP_ISOTOPE_ERROR	Misassigned ^{13}C	✔		0 to 2	MIS only
PEPMASS	Peptide mass		✔	>100	optionally followed by intensity and charge
PRECURSOR	Precursor m/z	✔		>100	
QUANTITATION	Quantitation method	✔		as defined in quantitation.xml	MIS only
RAWFILE	Raw file identifier		✔	string	MIS only

Tool	Formats
ProteoWizard	mzML, TraML, mzIdentML, mzXML, vendor formats
OpenMS	mzML, TraML, mzIdentML, mzData, mzQuantML, et al.
Trans-Proteomic Pipeline (TPP)	mzML, mzXML, pepXML, protXML (ProteoWizard)
compomics-utilities	MSF, tandem, mzML, omx, dat, FASTA
jmzReader	mzML, mzXML, mzData, PRIDE XML, dta, MGF, ms2, pkl
jTraML	TraML
multiplierz	Vendor formats
PEFF Viewer	PEFF
PRIDE Converter 2	mzTab, PRIDE XML (jmzReader)
Mascot & Distiller	MGF, mzML, mzXML, mzIdentML, vendor formats
SpectraST	msp, splib, blib, ASF, mzML, mzXML, pepXML, etc.
ProHits	PSI-MI (TPP formats)

Lecture 144

Open MS Data Formats

- Mass spectrometer outputs data with mass/charge ratios & respective ion intensities
- RAW file formats are specific to each instrument and each vendor has its own unique file format
- Once an instrument is upgraded, data output from the instrument is also changed
- Hence the underlying RAW file format needs to be upgraded as well

Need:

- Proprietary RAW formats are binary formats which are difficult to read and parse
- If you have the software from the maker of the MS then you can read the RAW data file as well

Tool	Formats
Anubis	TraML, mzML, mzXML
Proteios	TraML, mzML, mzXML
Skyline	.sky, .skyd, mzML, mzXML, vendor formats
ATAQS	TraML, mzML, mzXML
Corra	APML, mzXML
Java MIAPE API	PRIDE XML, mzML, mzIdentML, GelML

Conclusions:

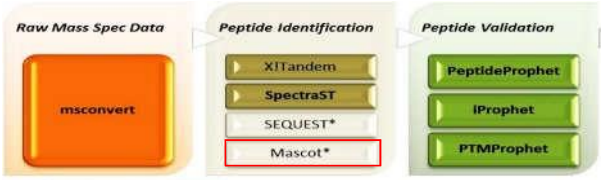
- Several software exist for converting RAW file formats into open software formats
- Each open format has its own unique advantages
- mzXML and MGF formats are most frequently used

Lecture 145

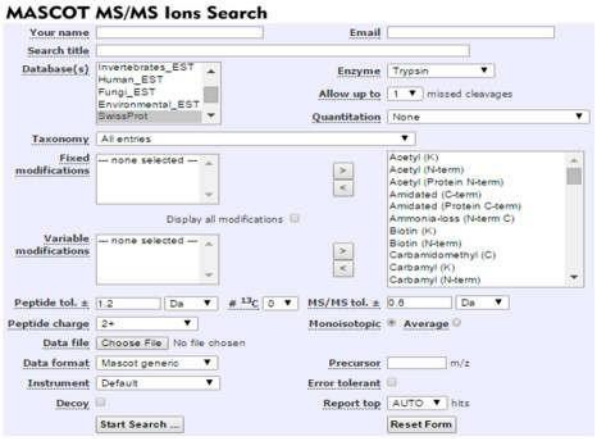
Online Proteomics Tools - MASCOT

Background:

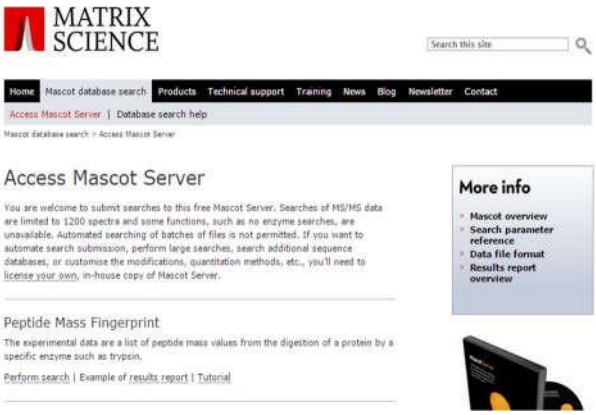
- Matrix Science developed an online Bottom Up Proteomics Search Engine
- "MASCOT"
- Mascot can search peptide mass fingerprinting and shotgun proteomics dataset



mzML http://tools.proteomecenter.org/software.php



http://www.matrixscience.com/search_form_select.html



Conclusions:

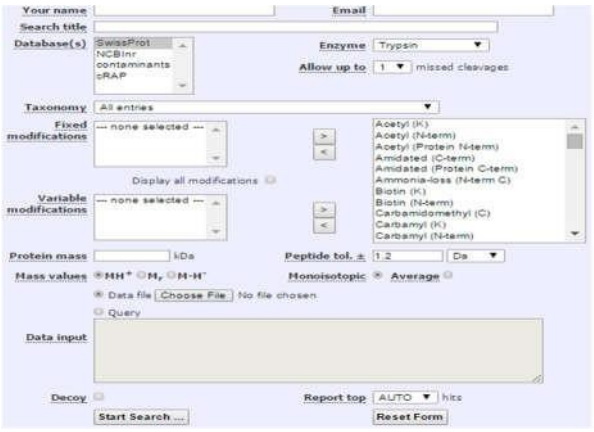
- Mascot is the most widely used online search tool for proteomics data
- However, it lacks a batch processing mode
- Also, it does not cater for top-down proteomics data

Lecture 146

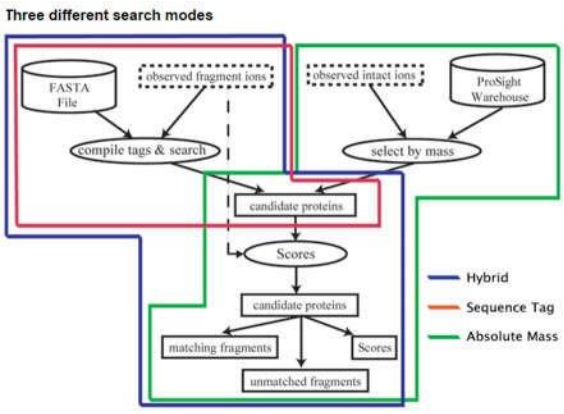
Online Proteomics Tools – ProSight PTM

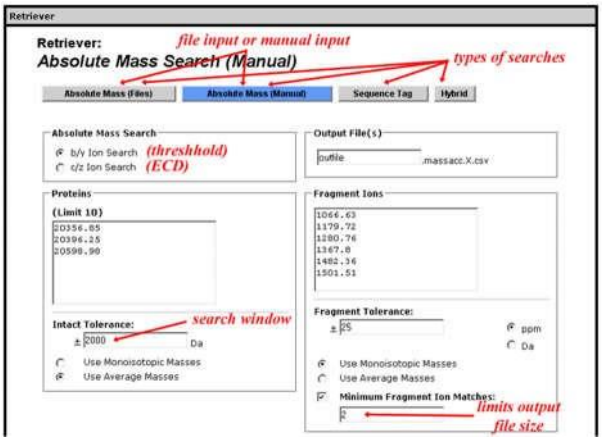
Background:

- Kelleher et al have developed an online Top Down Proteomics Search Engine
- "Prosight PTM"
- ProsightPTM searches top down proteomics data and reports the precursor protein



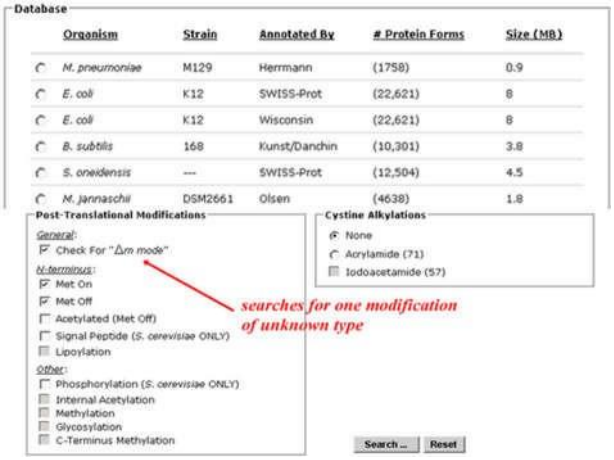
https://prosightptm.northwestern.edu/about_retriever.html





Isotopic variants of natural elements

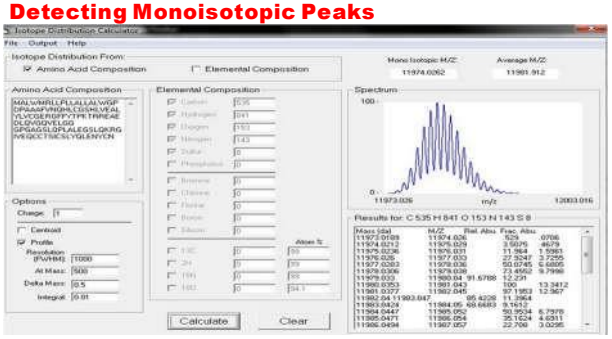
Element		Abundance (%)	Mass
Hydrogen	^{1}H	99.99	1.00783
	^{2}H	0.01	2.01410
Carbon	^{12}C	98.91	12.0000
	^{13}C	1.09	13.0034
Nitrogen	^{14}N	99.6	14.0031
	^{15}N	0.4	15.0001
Oxygen	^{16}O	99.76	15.9949
	^{17}O	0.04	16.9991
	^{18}O	0.20	17.9992
Phosphorus	^{31}P	100	30.9738
Sulfur	^{32}S	95.02	31.9721
	^{33}S	0.76	32.9715
	^{34}S	4.22	33.9676



Types of masses

- Nominal Mass
- Monoisotopic Mass
- Average Mass

Detecting Monoisotopic Peaks



Conclusions:

- ProSight PTM is the state of the art in top down proteomics search
- Using Prosight PTM, post-translational modifications can be accurately identified.

Lecture 147

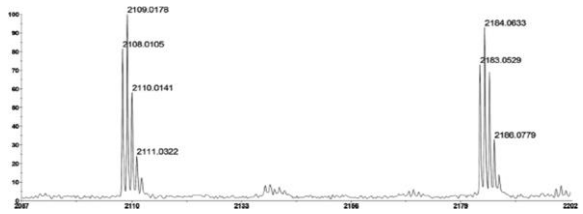
Example Case Study - I

- Proteomics is the study of protein sequence & structure
- Mass Spectrometry proteomics identifies as well as determines the protein sequence
- Case study of searching sample.

Step 1 – Monoisotopic Peak Detection

- Natural elements occur in multiple isotopes
- Isotopes differ in their masses
- The abundance of each isotopic variant is unique

Detecting Monoisotopic Peaks



Conclusions: MS1 data reports the isotopic distribution of intact molecule's mass

- Monoisotopic mass value has to be selected from this mass distribution
- This value is the highest mass value in the distribution

Lecture 148

Example Case Study - II

- The first step in protein identification and characterization using mass spectrometry involves intact protein/peptide mass measurement
- Next, we fragment the protein!
- A protein or peptide backbone may be fragmented any where along the peptide backbone
- This results in formation of two fragments i.e. N-term fragment and C-Term fragment

Possible Fragments:

- Let's take an example protein with 100 residues
- Such a molecule's backbone can be fragmented at 100 different locations
- The total number of possible fragments is then 200

Tandem MS (MS/MS):

- The mass of 200 fragments can then be measured by using an MS again
- The necessary condition for this measurement is that all 200 fragments are ionized.

Ionization Strategy for MS/MS:

- To ensure that all fragments of precursor molecule are also charged, we can use Electrospray ionization (ESI)
- ESI induces multiple charges on the intact molecule

Role of Electrospray Ionization

- Since ESI induces multiple charges on the precursor molecule, there is a good chance that upon precursors fragmentation, each fragment will have a portion of the charge.

ESI of an intact precursor and peptides

Conclusions:

- ESI allows for production of multiple charged ions
- Tandem MS helps measure molecular weights of ionized fragments.

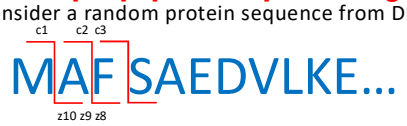
Lecture 149

Example Case Study - III

- Tandem MS helps measure the mass of the fragments
- How can we use this information to improve the search process for identification of intact proteins?
- Peptide sequence tags
- Those fragments which differ from each other by one amino acid's mass can provide clues on the sequence of proteins

Example peptide sequence tags

- Consider a random protein sequence from DB:



Conclusions:

- Peptide sequence tags help derive clues about the sequence of precursor proteins/peptides
- The short peptide sequences help us in shortlisting candidate proteins from the database.

Lecture 150

Example Case Study - IV

Background:

- MS1 helps measure the intact mass of proteins/peptides
- A list of candidate proteins/peptides can be formed by comparing MS1 mass to the mass of proteins/peptides in the database.
- MS2 or Tandem MS was performed after fragmentation of intact proteins
- MS2 helped extract peptide sequence tags from MS2 data
- Candidate proteins can be further shortlisted by PSTs

Introduction:

- Exhaustive matching of all MS2 peaks with the theoretical fragments of candidate proteins
- The set of theoretical fragments contains all possibilities of fragmentation.

Example theoretical and experimental fragments

- Consider a random protein sequence from DB:



Conclusions:

- Theoretical vs experimental fragments comparison helps as the third stage for shortlisting candidate protein list. This shortlisting will help you arrive at a small No. of proteins.

Lecture 151

Example Case Study - V

Background:

- MS1 & MS2 provide mass of intact molecules & its fragments
- This information helps filter proteins from protein database
- For a quantitative measure, scoring scheme is required

Example intact protein mass score

- Consider a random protein sequence from DB:

MAFSAEDVLKE…

$$\text{Score}_{M} = \frac{1}{\sqrt{(M_{Exp} - MT)^2}}$$

Example peptide sequence tags

- Consider a random protein sequence from DB:

MAFSAEDVLKE… n

$$PST_{Score} = \sum_{i=0}^{n} \left(\frac{Length(PSTi)}{RMSE_i}\right)$$

Example theoretical and experimental mass

- Consider a random protein sequence from DB:

MAFSAEDVLKE…

Conclusions:

- Three scoring schemes can be applied to score the match at each stage of protein search
- These scoring elements can be integrated to arrive at an over all candidate protein score!

Lecture 152

Example Case Study - VI

Background:

- Comparisons can be performed at various levels of information
- These include MS1, MS2, PSTs and theoretical fragments comparison
- Integrated scoring schemes couple these factors

Example basic scoring scheme

Simply sum the scores up (a linear function)

Example comprehensive scoring scheme

$$\text{Score} = \text{Score}_{MW} + \text{Score}_{PST} + \text{Score}_{Exp <> TMr}$$

$$\text{Score} = \frac{\text{Score}_{MW}}{} + \sum^{m} \frac{\text{Score}_{PST}}{} + \sum^{n} \frac{\text{Score}_{Exp <> TMr}}{}$$

Conclusions:

- A comprehensive scoring scheme can combine all the scores
- Several optimizations can be undertaken on the scoring scheme to further improve protein identification

Lecture 153

Properties of Amino Acids - I

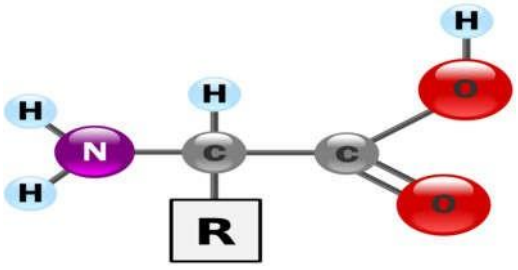
Background:

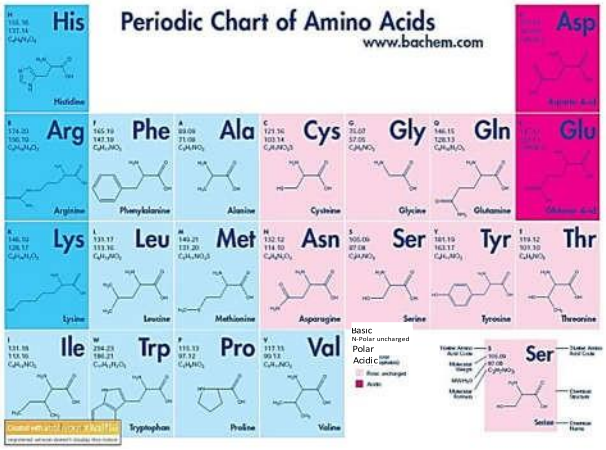
- Proteins are assembled by Ribosomes in cellular cytoplasm
- Each protein is synthesized by polymerization of amino acids
- Proteins take properties from the constituent amino acids

Introduction:

- There are 20 different amino acids
- Each has a different chemical composition giving rise to unique properties
- Structural commonalities exist between them

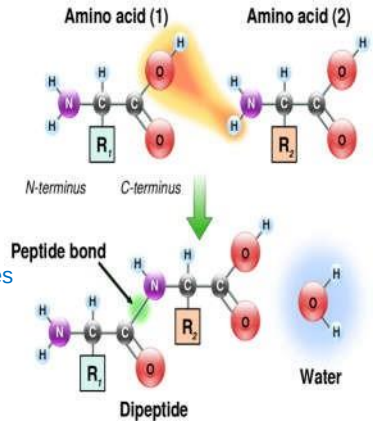
Chemical Structure of an Amino Acid





$$E_{MW} \quad {}_{i=O}\,RMSE_{PST} \quad i=0 \quad E_{EXP\ <>\ TMr}$$

Polymerization of Amino Acids



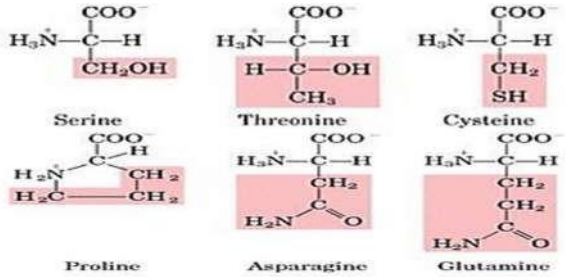
Conclusions:

- Amino acids have unique properties such as polarity, charge states and interactions with water
- Each of these properties describes the over all characteristic of an amino acid

Polar Amino Acids

- These amino acids are polar but are not charged i.e. no net charge on the amino acid
- Prefer to reside / interact with aqueous environments
- Mostly found at the surface of folded proteins

Polar Amino Acids



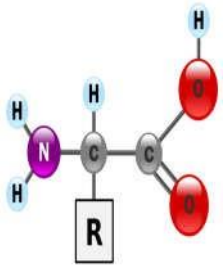
Lecture 154

Properties of Amino Acids - II

Background: Amino acids have characteristics like polarity, hydrophobicity, and charge states

These characteristics are governed by the elemental composition of an amino acid's side chain (R group)

R Group in an Amino Acid



Hydrophobic Amino Acids:

Since H and C introduce very little dipole moments in hydrophobic amino acids, these amino acids are non polar. Hydrophobic amino acids are mostly found at the inside of folded proteins

Conclusions:

- Amino acids have unique properties such as polarity, charge states and interactions with water
- Each of these properties describes the over all characteristic of an amino acid.

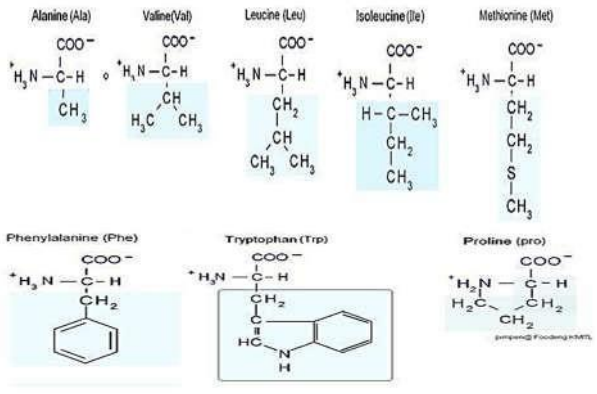
Lecture 155

Properties of Amino Acids - III

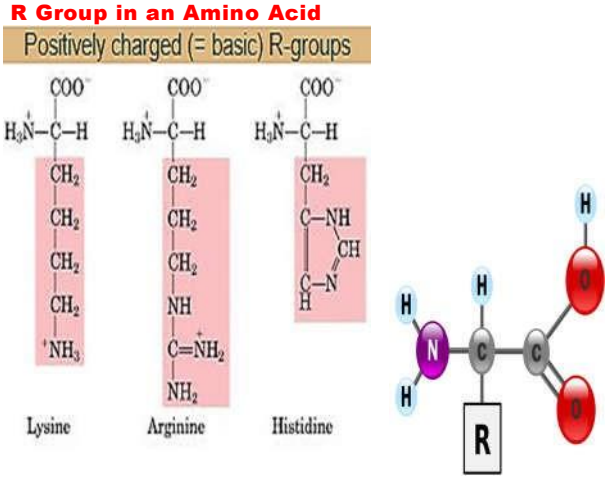
Background:

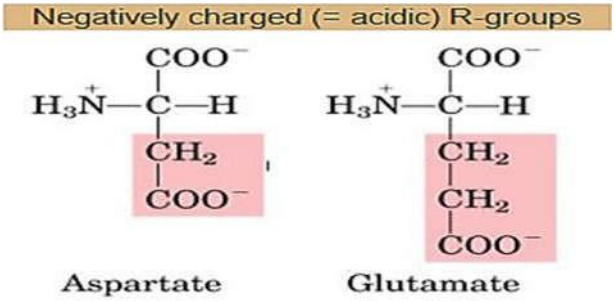
- Amino acids have characteristics like polarity, hydrophobicity, and charge states
- These characteristics are governed by the elemental composition of an amino acid's side chain (R group)

Hydrophobic Amino Acids contain chains of C and H in their R group:



R Group in an Amino Acid



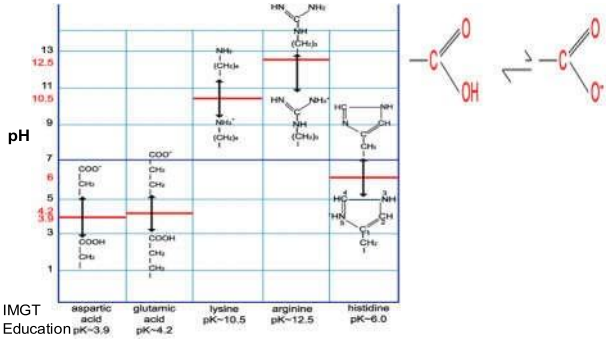


Change in Charge:

- Upon polymerization of amino acids into polypeptide chains, charged amino acids get neutralized
- At pH=7, five amino acids are charged, 2 negatively and 3 positively

Charge & pH

- If pH > pK for an amino acid, the carboxyl side chains loses a proton (H+) and become negatively charged, hence acidic.



Conclusions:

- Five amino acids have unique charge states and are hence eager for interactions
- These amino acids are typically found on the surface of proteins.

Lecture 156

Properties of Amino Acids - IV

Background:

- Some amino acids may be positively or negatively charged
- The charge state depends on the acceptance or donation of protons from the R group side chains

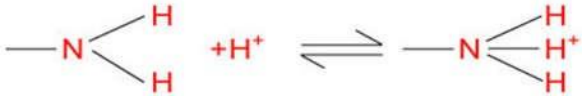
pK of charged amino acids

- pK values for an amino acid is the pH at which exactly half of the chargeable group is charged.

Charge & pH

- If pH < pK for an amino acid, the amine side chains gain a proton (H+) and become positively charged, hence basic.

	Amino acid	pK of the side chain group
-ve	Aspartic acid	3.9
	Glutamic acid	4.2
+ve	Lysine	10.5
	Arginine	12.5
	Histidine	6.0



Conclusions:

- Depending on the pH, an amino acid may become charged
- This may be positive or negative depending on the amino acid.

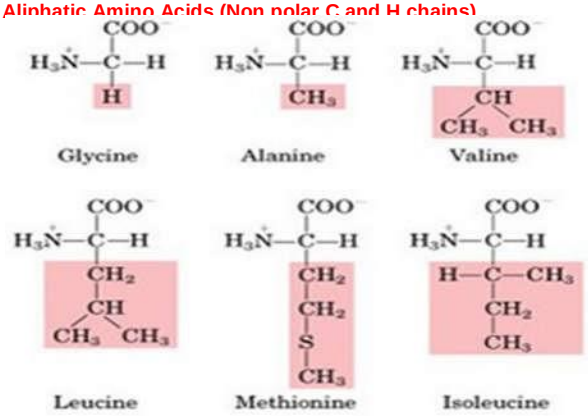
Lecture 157

Properties of Amino Acids - V

Background:

- Amino acids may be charged depending on pH
- This depends on the charge acceptance or donation from within an amino acid
- Additionally, amino acids have structures as well

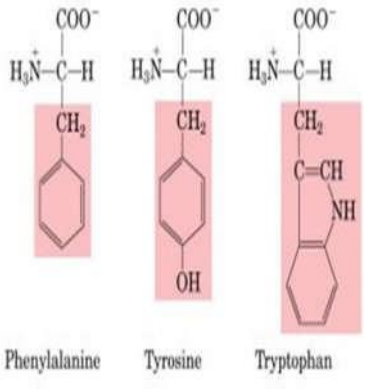
Aliphatic Amino Acids (Non polar C and H chains)



Properties imparted by side chains:

- Side chains comprising merely of Carbon and Hydrogen are:
  - Chemically inert,
  - Poorly soluble in water.

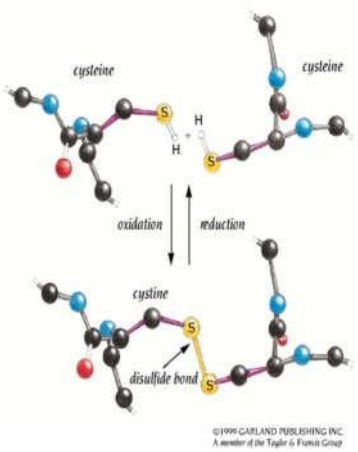
Aromatic R groups



Example 3: Cystine

Cysteines cement together by making disulfide bonds to stabilize 3-D protein structures

In eukaryotes, disulfide bonds can be found in secreted proteins or extracellular domains



- However, side chains containing organic acids are very different.
- They are chemically reactive
- Soluble in water

Conclusions:

- Elemental composition plays a very important role in determining properties of amino acids
- Solubility and reactivity are key factors participating in protein folding.

Lecture 158

Structural Traits of Amino Acids - I

Amino acids have several properties such as charge state, polarity and hydrophobicity

It is important to note that the physical size of each amino acid also varies

Effects of variation in amino acid size?

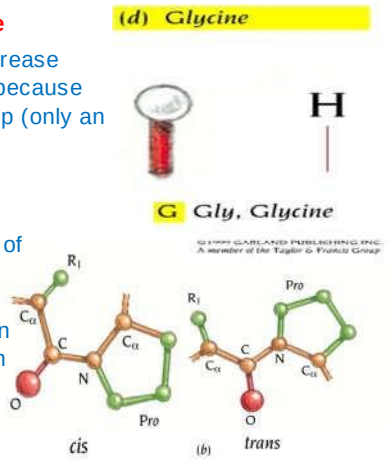
Example 1: Glycine

Glycine residues increase backbone flexibility because they have no R group (only an H), hence agile!

Example 2: Proline

Proline residues reduce the flexibility of polypeptide chains!

Proline cis-trans isomerization is often a rate-limiting step in protein folding



Conclusions:

- Amino acids not only have physical and chemical properties, but also structural properties
- These structural properties are equally important in giving rise to protein structures

Lecture 159

Structural Traits of Amino Acids - II

Background:

- Each amino acid has a unique set of properties such as charge state, polarity and hydrophobicity
- Moreover, it may have unique structural traits as well which can help in protein folding

Introduction:

- Since some amino acids are hydrophobic, they may be employed in forming a stable core in a protein
- Also, chemically inactive amino acids reduce chances of destabilizing reactions in core

Packing of Hydrophobic Residues:

Packing/ burying of hydrophobic side chains into interior is a key driving force in protein folding

Problem in burying hydrophobic amino acids in protein core:

- Backbone is highly polar (hydrophilic) due to polar -NH and C=O in each peptide unit; these polar groups must be neutralized!

Solution: Form regular secondary structures!
Such as:

- Alpha Helices
- Beta Sheets which are stabilized by H-bonds!

Lecture 160

Structural Traits of Amino Acids - III

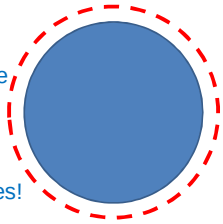
- The size and structure of each amino acid is unique
- Coupled with their chemical properties, each amino acid can uniquely contribute in the protein folding process.

Hydrophobic core formed by packed secondary structural elements provides compact, stable core
Upon establishment of a stable protein core, unstable or reactive groups can be added!

Addition of functional groups: "Functional groups" of protein are attached to the hydrophobic core framework. Surface or a protein or its exterior must have more flexible regions (loops) and polar/charged residues

- The very few hydrophobic "patches" on protein surface are involved in protein-protein interactions
- The active regions in a protein are almost all present on the surface!

Organization of core & surface in a protein



Conclusion: Each component of the protein structure has a unique and precise role in the construction of proteins Hydrophobic & hydrophilic components have equally useful roles!

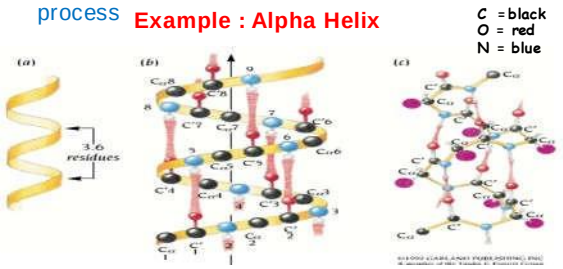
Lecture 161

Structural Traits of Amino Acids - IV

Background:

- The size and structure of each amino acid is unique
- Coupled with their chemical properties, each amino acid can uniquely contribute in the protein folding process

Example : Alpha Helix



Conclusion:

- Alpha Helix is an example of amino acid folding
- Stabilized by H-bonds between every ~ 4th residue in backbone
- Reactive amino acids are exposed for external interactions

Lecture 162

Introduction to Protein Folding

- Proteins comprise 20 different amino acids
- Amino acids polymerize and form protein molecules
- Proteins fold together to take 3D forms
- But how does a protein actually fold?
- The answer is still unknown!
- Scientists have spent decades in trying to find a definite answer to this question, but to no avail

Folding of Proteins:

- After polymerization of amino acids, linear chains are formed
- When these chains of amino acids are put in water, the proteins fold spontaneously!

How do we know the final folding state of a protein?

- The folded protein molecule should have the lowest possible energy !
- Anfinsen's dogma
- Unique, stable & kinetically accessible minimum free energy.

Conclusion: Proteins fold spontaneously. Proteins fold to achieve thermodynamic stability. Proteins fold to organize themselves for performing functions in cells.

Overall Energy (stability) of the Protein

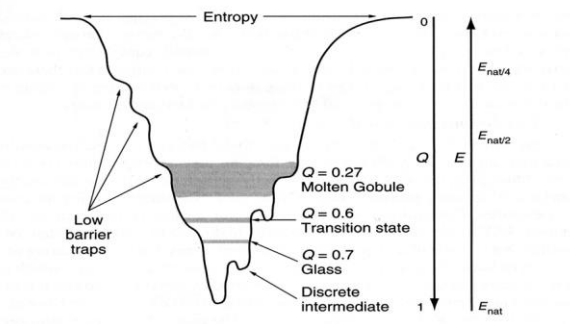


Figure 19.16 Cross section through a folding funnel. *E* corresponds to free energy. [Courtesy of P. G. Wolynes]

Lecture 163

Importance of Protein Folding

Background:

- Proteins fold spontaneously
- Proteins fold to achieve thermodynamic stability
- Proteins fold to organize themselves for performing functions in cells.

Why study folding

- Proteins are the functional machines in cells
- Dysregulated protein expressions are a major cause of disease
- Understanding protein folding helps design suitable drugs

Size of the problem?

- If each amino acid can take 3 different conformations, and its parent protein has 100 amino acids, then what are the number of combinations?
- $100^3 = 5 \times 10^{47}$. A huge number of combinations!
- Given 5×10^{47} combinations, how much time is needed to fold this protein? If it take $1/10^{th}$ of a nano-second(10^{-10}), then to compute all the folding possibilities will take 1.6×10^{30} years.

Real time folding

- With so much time required to fold a protein, how do the proteins fold then?
- In fact, it take a protein less than a second to fold!!!
- Amazing speed of folding!
- How do proteins do that?

Computational folding of proteins:

- If a protein is misfolded, then it can lead to a lack of function in the protein
- To study anomalies in structures & to discover newer structural forms, computational algorithms are used
- How do we study folding, computationally?
- First, we collect clues & evidences from experimentally reported structures
- We utilize these observations to analyze unknown structures.

Conclusions:

- Given algorithms and procedures to fold a protein, we can fold amino acid chains to form 3D proteins
- This can help us study misfoldings, interactions between drugs and proteins etc.

Overall Energy (stability) of the Protein

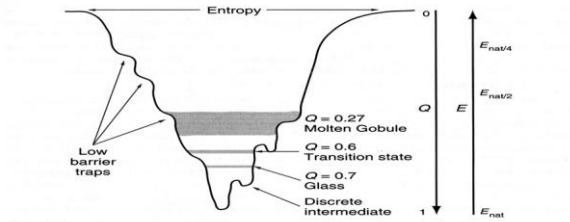


Figure 19.16 Cross section through a folding funnel. *E* corresponds to free energy. [Courtesy of P. G. Wolynes]

Conclusion:

- This is called "Levinthal's Paradox"
- We will try to understand this folding process using experimental datasets and algorithms
- Molecular simulations are also helpful for it!

Lecture 164

Computing Protein Folding Possibilities

Background:

- Computing the protein folding can help us study mis-folding, interaction between drugs and proteins etc.
- However, first, it is important to know the number of the protein folding possibilities!

Introduction:

- Let's assume that each amino acid can fold into three different conformations
- They are Alpha Helices, Beta Sheets and Loops
- We know that proteins comprise of 100s of amino acids.

Lecture 165

Process of Protein Folding

- Levinthal's Paradox- enormous time required to compute all folding possibilities. It's impossible to consider all the possibilities computationally.
- So, we are trying to understand the folding process

The forces involved in protein folding include:

- Electrostatic interactions
- van der Waals interactions
- Hydrogen bonds
- Hydrophobic interactions

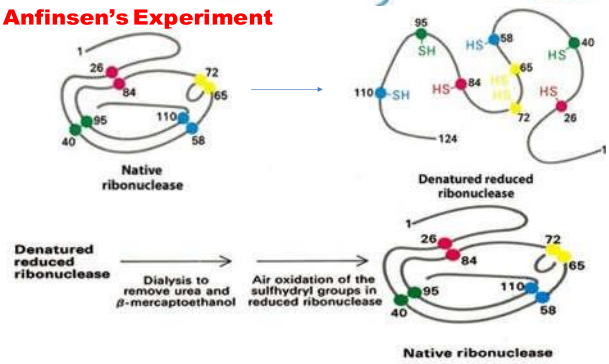
Conclusion: All the information required for folding a protein into its native structure is present within the protein's amino acid sequence!

The native folded form of protein is thermodynamically most stable as compared to others.

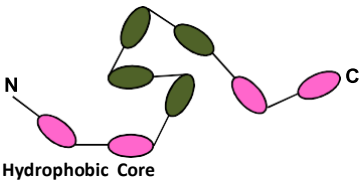
Example



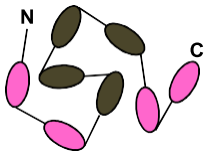
Anfinsen's Experiment



Nuclear Condensation Model



Step 1: Formation of a Hydrophobic Core



Step 2: Including remaining amino acids and expanding the nucleus

Lecture 166

Models of Protein Folding

Background

- Information required for folding a protein into its native structure is present within the protein's amino acid sequence!
- The native folded form of protein is thermodynamically most stable as compared to others.

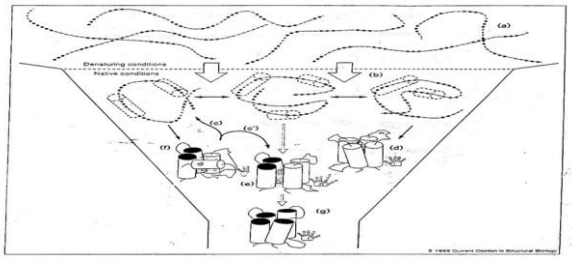
Conclusion:

- Several models exist for folding a protein given its amino acid sequence
- The fundamental requirement is that the folding process remain spontaneous
- There is still no definitive folding hypothesis!

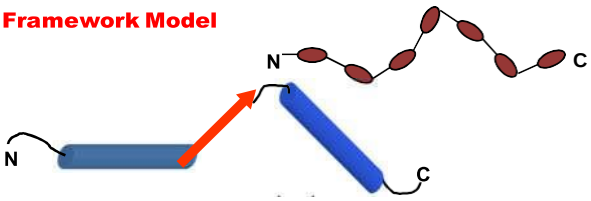
Lecture 167

Protein Structures

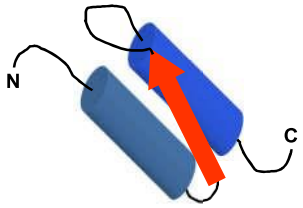
- Proteins spontaneously fold to take 3D forms
- It's a fast yet specific process which leads to a folded protein
- Several forces act together to fold the protein structure



Framework Model



Step 1: Formation of secondary structures

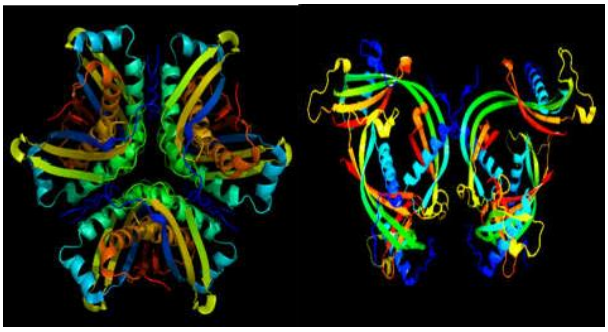


Step 2: Arrangement of secondary structures

Energies of Various Bonds & Interactions

Bond Type	kJ/mol
Covalent Bond	250
Electrostatic	5
van der Waals	5
Hydrogen bond	20

Hydroperoxide resistance protein OsmC (1vla) Cystatin – 3 (C)



Conclusion: Protein structures are very complex yet they form spontaneously. We will investigate how to develop algorithms to predict such structures!

- 3' structures represent the complete monomeric protein structure
- 3'structures can combine with other polypeptide units to form a quaternary structure.

- Quaternary structures are also called 4' structures
- 4' structures are exemplified by protein complexes etc.

Conclusion:
- Protein structures are organized into 1', 2', 3' and 4' modular conformations.
- We will investigate how to develop algorithms to predict these structures!

Lecture 168

Primary, Secondary, Tertiary and Quaternary Structures:

Background
- Complex protein structures form spontaneously as a protein folds.
- A huge variety of protein structures exist
- Each structure is designed to perform a specific function.

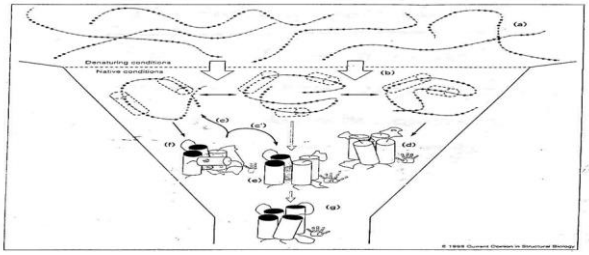
Introduction
- Interestingly, each protein mega structure gets built out of only a few sub-structures!
- Combinations from the SMALL substructure set are used to construct larger protein structures

Types of structures
- Single Alphabet Amino acid tags can be put together linearly to represent a protein sequence
- This sequence is also called the primary sequence
- Primary sequence can also be referred to as 1' structure
- Sub-structures are formed as a result of 1' structure's folding
- Folded sub structures are called secondary protein structures. 2ndry structures are also referred to as 2' structures
- 2' sub-structures are packed together to form super structures
- These protein super structures are called tertiary structures
- Tertiary structures are also referred to as 3' structures

Lecture 169

Primary Structure of Proteins

- Protein structures are organized into 1', 2', 3' and 4' modular conformations
- 1'structures are essentially the amino acid sequence of the proteins.



Amino Acid	3-Letter Code	1-Letter Code
Alanine	Ala	A
Cysteine	Cys	C
Aspartic acid or aspartate	Asp	D
Glutamic acid or glutamate	Glu	E
Phenylalanine	Phe	F
Glycine	Gly	G
Histidine	His	H
Isoleucine	Ile	I
Lysine	Lys	K
Leucine	Leu	L
Methionine	Met	M
Asparagine	Asn	N
Proline	Pro	P
Glutamine	Gln	Q
Arginine	Arg	R
Serine	Ser	S
Threonine	Thr	T
Valine	Val	V
Tryptophan	Trp	W
Tyrosine	Tyr	Y

Methods for obtaining 1' structures

- Edman Degradation
- Tandem Mass Spectrometry.

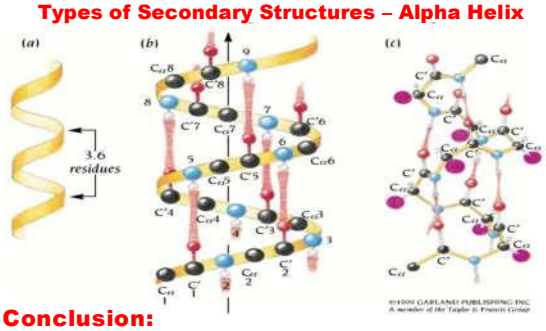
Databases

- 1'structure databases are essentially protein sequence databases
- Examples include Uniprot, Swissprot amongst several others.

Conclusion:

- Protein sequences are the primary structures of proteins
- The primary or 1' structure of a protein determines its initial properties
- 1' structure lays the foundation for 2' structures.

Types of Secondary Structures – Alpha Helix



Conclusion:

- Protein sequences fold onto themselves and make H-Bonds to create 2' structures
- Several types of 2' structures exist
- These include Alpha Helices and Beta Sheets

Lecture 170

Secondary Structures of Proteins - I

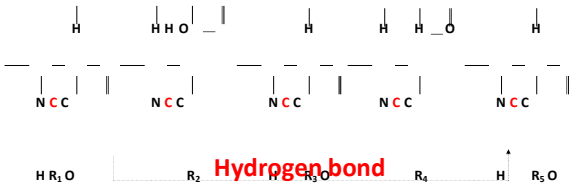
Introduction

- The primary or 1' structure of a protein determines its basic properties
- 1' structure lays the foundation for 2' structures
- 2' structures are also referred to as secondary structures.

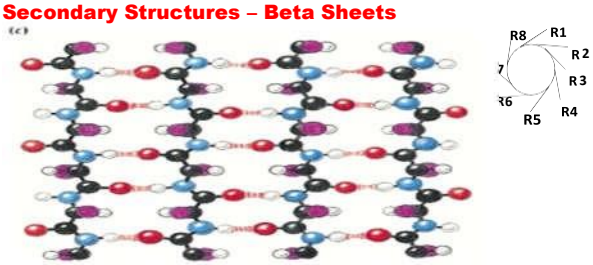
Formation of 2' structure

- C- terminus is negatively charged
- N-terminus is positively charged
- C and N termini can therefore make Hydrogen Bonds
- Hydrogen Bonds are the reason of 2' structure formation

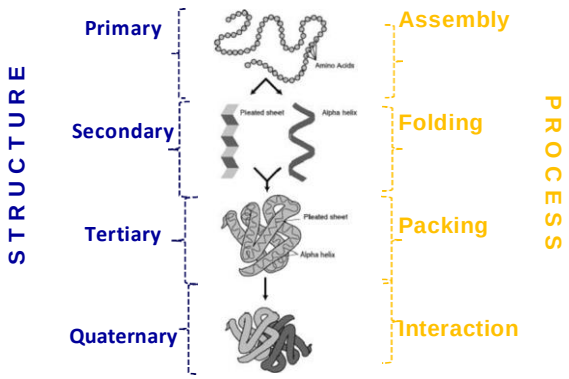
Forming Secondary Structure



Secondary Structures – Beta Sheets



Organization of Secondary Structure



Lecture 171

Secondary Structures of Proteins - II

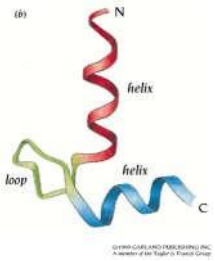
- 2' structures or secondary protein structures are formed as a result of H-Bond formation between N and C termini in a protein backbone
- Types of 2' structures include Alpha helices & Beta sheets

Properties of Loop

- Loops connect helices and sheets
- Loops vary in length and 3-D configurations
- Loops are mostly located on the surface of proteins
- Loops are more "acceptable" of mutations
- Loops are flexible & can adopt multiple conformations
- Loops tend to have charged and polar amino acids
- Loops are frequently components of active sites

A Special Secondary Structure

Coils



- Secondary structure that are not helices, sheets, or recognizable turns
- Disordered regions, but also appear to play important functional roles

Conclusion:

- Loops and Coils are also secondary structure which form the first structures after folding of protein's amino acids
- Loops and Coils are very important 2' structures in that they form active sites of proteins.

Lecture 173

Quaternary Structures of Proteins

- 4' structures or quaternary structures are formed by different peptide chains that make up the protein
- Multimeric proteins which comprise of multiple peptides form 4' structures

Monomeric vs. Multimeric Proteins

- Protein comprised of only a single chain (monomeric) donot have a quaternary structure
- Proteins with multiple chains can form 4' structures.

Conclusion:

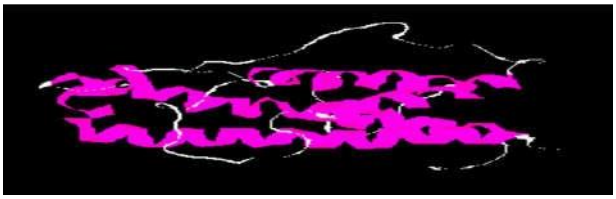
- 4' structures are kept in conformation by Hydrogen Bonds, Covalent Disulphide Bonds, Hydrophobic Interactions and ionic bonds
- In terms of stability 4' > 3' > 2' > 1'

Lecture 172

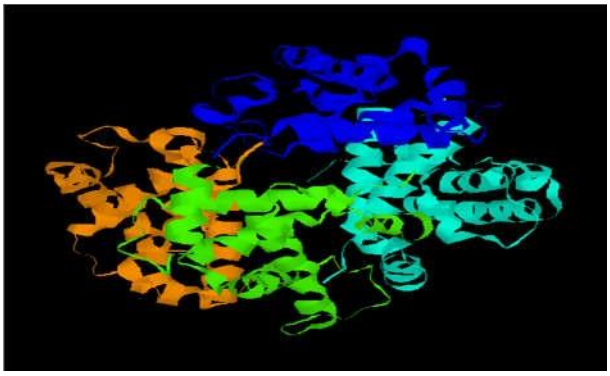
Tertiary Structures of Proteins

- 2' structures include alpha helices, beta sheets, loops and coils
- Upon combination of 2' structures, a tertiary or 3' structure is formed
- 3' structure is next level of structure organization

Example of Tertiary Structure Human Growth Hormone.



Example of Quaternary Structure
See how 2' and 3' structures come together!



Formation of 3' structure

- Hydrophobic interactions between nonpolar R-groups
- Covalent bonds in the form of Disulphide bridges

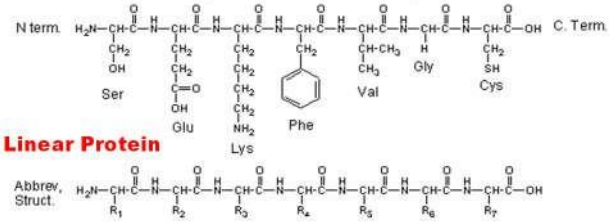
Conclusion:

- Combinations of Alpha helices, Beta sheets, coils and loops help form 3' structures
- Covalent bonds, Hydrogen bonds and hydrophobic interactions enforce the 3' structure

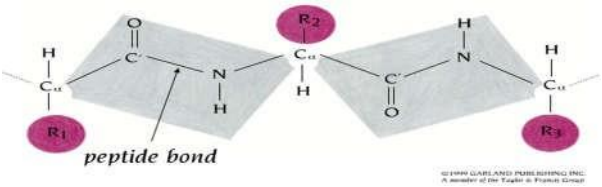
Lecture 174

Introduction to Bond Angles in Proteins
Introduction

- Protein folding results in a linear chain of amino acids getting packed into a compact 3D structure
- This leads to a reduction in bond angles from an initial of 180 degrees (protein's linear form).



Formation of Planar Peptide Bond

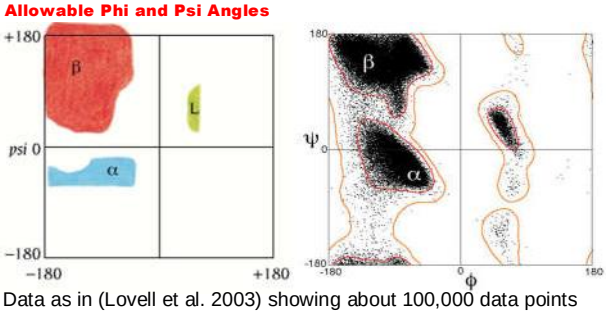


1. The resultant chain gets its own set of attributes
2. Peptide bond is planar & rigid!

Dihedral Angles

- Angle between two planes (i.e. 4 points)!
- Considering the middle two points to be aligned (or overlapped), the angle between the 1st, overlapped and the 4th points forms a dihedral angle.

Allowable Phi and Psi Angles

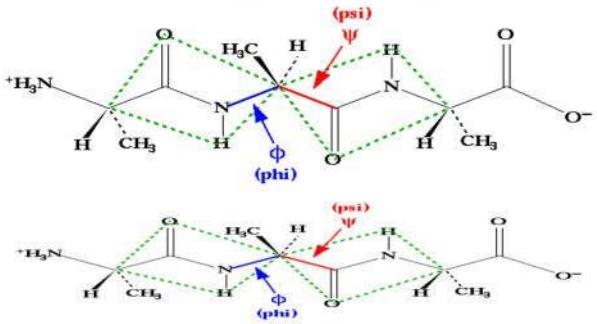


Data as in (Lovell et al. 2003) showing about 100,000 data points for several amino-acids

Conclusion: A limited range of Phi and psi angles are taken up as a result of folding

- This range of angles constitutes the allowable range of torsion/rotation angles that are taken up by the protein

Protein After Folding: Phi and Psi Angles



ϕ (*phi*, involving C'-N-Cᵅ-C')

ψ (*psi*, involving N-Cᵅ-C'-N)

Lecture 176

Structure Visualization - I

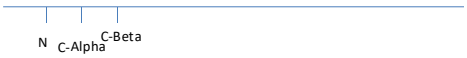
Introduction

- We know that protein backbone takes up specific rotation angles after folding
- Given the exact torsion angles, can we reconstruct a protein structure using computational tools?

What exactly do we visualize?

- A protein consists of multiple amino acids
- Each amino acid has a C-terminus and an N-Terminus
- The protein is also in a complex 3D folded form, so what to begin with?

Protein Backbone and C atoms



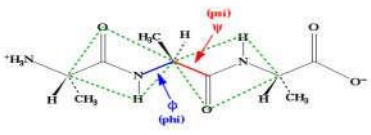
Conclusion:

- Proteins fold into 3D structures
- Phi and psi angles are taken up as a result of folding
- These angles can be measured towards understanding the protein structure.

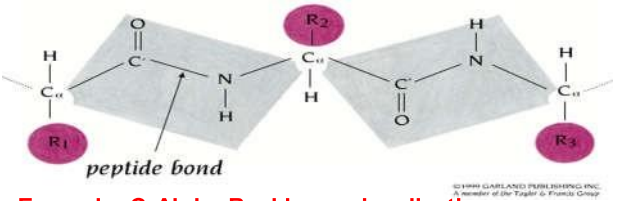
Lecture 175

Ramachandran Plot

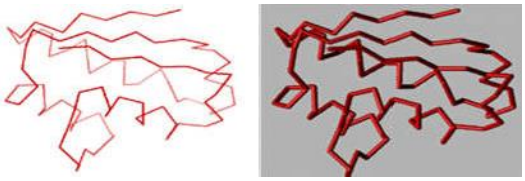
- Phi and Psi angles can be measured with in the folded structures
- ϕ - *phi*
- Involves C'-N-Cᵅ-C'
- ψ – *psi*
- Involves N-Cᵅ-C'-N



Omitting Planar bonds and Tracing C-Alpha atoms in backbone



Example: C-Alpha Backbone visualization



Conclusion:

- C-Alpha atoms are traced to recreate a 3D protein structure
- The choice is made while keeping planar nature of the peptide bond in view
- Later we will see how to insert side chains into the visual models as well.

Lecture 177

Structure Visualization - II

Introduction

- C-Alphas can be used to construct the backbone of a protein towards its visualization
- We also need a representation of measurements for assigning the atomic distances!

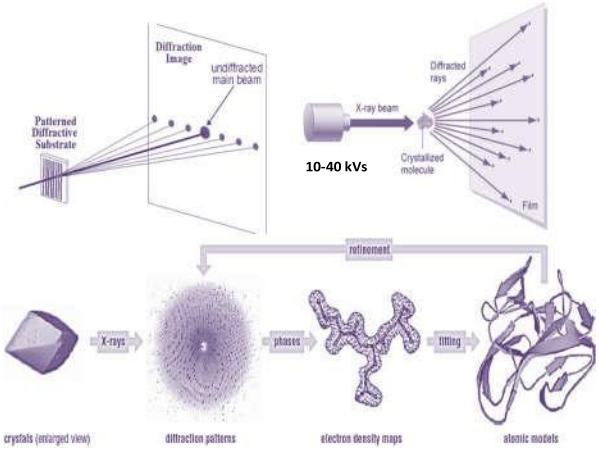


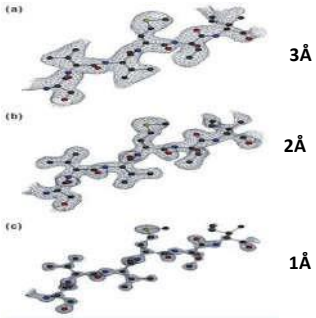
Ansedel Anders Ångström (1814–1874)

The ångström is used to express the size of atoms, molecules and extremely **small biological structures**, the **lengths of chemical bonds**, the **arrangement of atoms in crystals**.

1 **angstrom** is a unit of length equal to 10^{-10} m (one ten-billionth of a meter) or 0.1 nm

Atoms of phosphorus, sulfur, and chlorine are ~1 Å in covalent radius, while a hydrogen atom is 0.25 Å





Lower Resolution

3Å

2Å

Higher Resolution

1Å

Conclusion:

- C-Alpha atoms are traced to recreate a 3D protein structure
- Each C-Alpha atom is at a distance which can be represented in the unit "Angstrom"
- 1 A resolution is better than 10 A

Lecture 178

Experimental Determination of Protein Structure:

Introduction:

- C-Alpha atoms are traced to recreate a 3D protein structure
- Distances between C-Alphas are measured in the unit "Angstrom"
- So, how do we measure these distances?

X-Ray Crystallography

- Crystallography data gives relative positions of atomic coordinates
- The data is obtained from diffractions by the atoms in a protein structure
- The coordinates of each atom in x,y and z axis are output!

Conclusion:

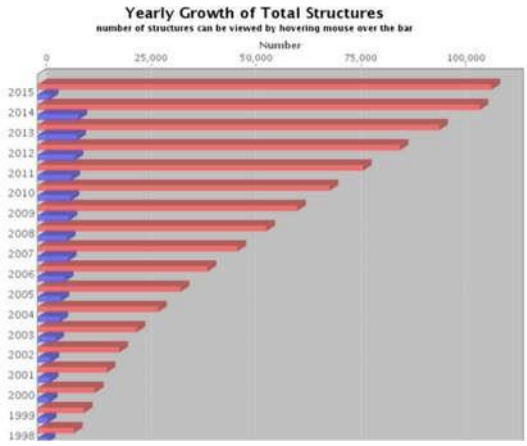
- Crystallized proteins are used to determine protein structures.
- As X-rays diffract from the atoms in a protein, the atomic distances are noted.
- These distance in 3D are measured in Angstroms.

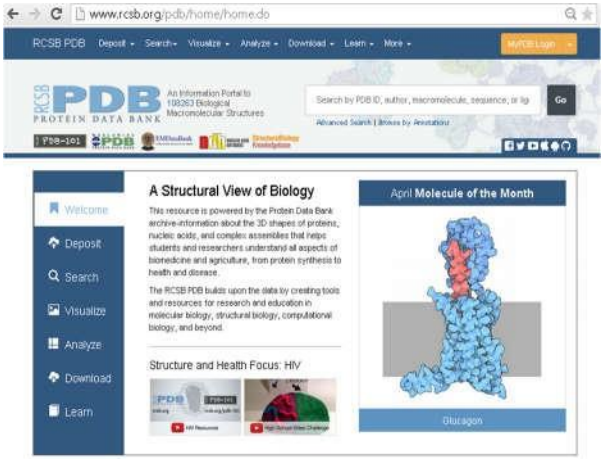
Lecture 179

Protein Databank

Introduction:

- Position of of C-Alpha atoms are used to construct 3D protein structure
- X-Ray diffraction data helps measure the atomic positions
- X, Y & Z positions of several proteins are available online





PDB File Format

HEADER – Contains a brief description of the structure, the date and the PDB ID code.

TITLE – The title of the structure.

COMPND – Brief details of the structure.

SOURCE – Identifies which organism the structure came from.

KEYWDS – Lists a set of useful words/phrases that describe the structure.

AUTHOR – The scientists depositing the structure.

REVDAT – The date of the last revision.

JRNL – One or more literature references that describe the structure.

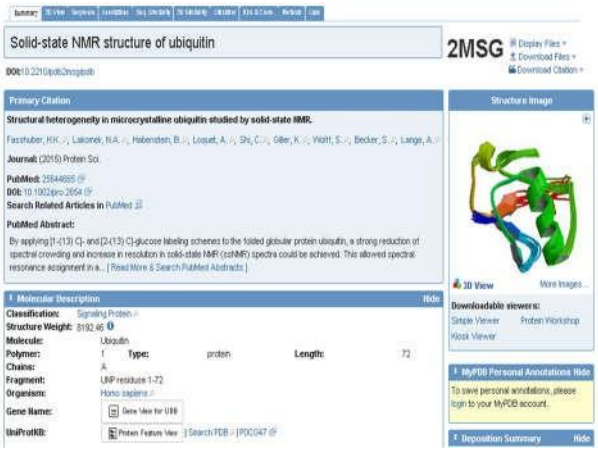
REMARK 1 through REMARK 999 – Details of the experimental methods used to determine the structure are contained in this subsection (see the example in the next section).

DBREF – Cross links to other databases.

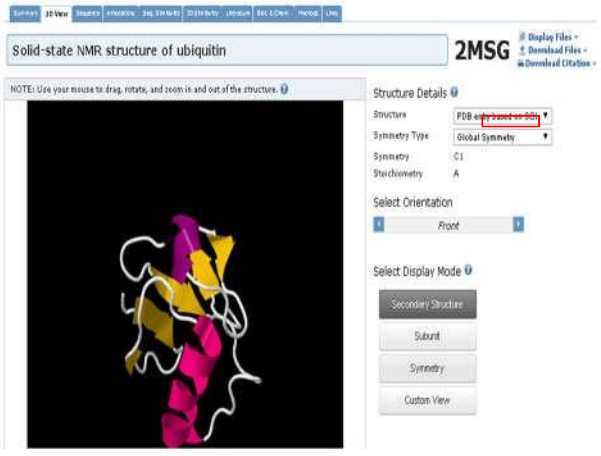
SEQRES – The official amino acid sequence (protein, RNA or DNA) of the structure.

HELIX/SHEET – Details of the regions of secondary structure found in the protein.

ATOM/HETATM – The 3D spatial coordinates of particular atoms in the protein structure (the "ATOM" lines) or other molecules such as water or co-factors (the "HETATM" lines).




```
ATOM      1  N   MET A   1     102.329 111.862  92.452  1.00 78.64           N
ATOM      2  CA  MET A   1     103.332 112.165  93.516  1.00 77.39           C
ATOM      3  C   MET A   1     103.877 113.584  93.255  1.00 76.87           C
ATOM      4  O   MET A   1     103.802 114.075  92.129  1.00 78.54           O
ATOM      5  CB  MET A   1     104.437 111.099  93.495  1.00 78.51           C
ATOM      6  CG  MET A   1     105.176 110.881  94.812  1.00 76.25           C
ATOM      7  SD  MET A   1     106.505 112.076  95.116  1.00 76.95           S
ATOM      8  CE  MET A   1     107.077 111.551  96.763  1.00 73.31           C
ATOM      9  N   GLU A   2     104.404 114.235  94.292  1.00 74.31           N
ATOM     10  CA  GLU A   2     104.917 115.609  94.211  1.00 70.70           C
ATOM     11  C   GLU A   2     105.963 115.909  93.143  1.00 68.10           C
ATOM     12  O   GLU A   2     106.048 117.044  92.683  1.00 67.95           O
ATOM     13  CB  GLU A   2     105.464 116.053  95.574  1.00 75.97           C
ATOM     14  CG  GLU A   2     106.692 115.246  96.029  1.00 82.57           C
ATOM     15  CD  GLU A   2     107.263 115.682  97.378  1.00 83.76           C
ATOM     16  OE1 GLU A   2     106.611 115.396  98.412  1.00 86.90           O
ATOM     17  OE2 GLU A   2     108.373 116.276  97.401  1.00 81.23           O
ATOM     18  N   ASN A   3     106.789 114.924  92.784  1.00 64.50           N
ATOM     19  CA  ASN A   3     107.834 115.134  91.773  1.00 59.93           C
ATOM     20  C   ASN A   3     107.381 114.767  90.360  1.00 55.41           C
ATOM     21  O   ASN A   3     108.159 114.856  89.416  1.00 53.33           O
ATOM     22  CB  ASN A   3     109.086 114.308  92.095  1.00 62.56           C
ATOM     23  CG  ASN A   3     109.531 114.441  93.535  1.00 62.42           C
ATOM     24  OD1 ASN A   3     109.045 113.724  94.408  1.00 63.06           O
ATOM     25  ND2 ASN A   3     110.484 115.326  93.787  1.00 62.31           N
```



Conclusion:

- PDB contains protein structure information
- It has the coordinates of C-Alphas for over 50,000 proteins
- Protein structures can be visualized using this information

Lecture 180

Visualization Technique

Background

- Proteins fold into 3D structures
- Phi and psi angles are assumed as a result of folding
- These angles can be measured and viewed towards understanding the protein structure

Introduction

- To view a protein, we need to evaluate the physical location of its atoms
- Proteins have Carbon and Nitrogen in their backbone
- How can we proceed to view these atoms?

Protein Backbone, Carbon-Alpha (CA)

CA atomic coordinates

To trace the backbone of a protein, CA atoms trace can be used. Note that CA atoms have the side chains attached to them.

CA coordinates can be found in the PDB file

Conclusion:

Protein structures can be visualized by tracing the CA atoms

Coordinates of CA atoms can be obtained from the PDB

- Next, we need a tool to plot these coordinates

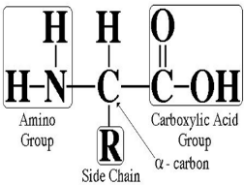
ATOM	1	N	MET	A	1	102.329	111.862	92.452	1.00	78.64	N
ATOM	2	CA	MET	A	1	103.332	112.165	93.516	1.00	77.39	C
ATOM	3	C	MET	A	1	103.877	113.584	93.255	1.00	76.87	C
ATOM	4	O	MET	A	1	103.802	114.075	92.129	1.00	78.54	O
ATOM	5	CB	MET	A	1	104.437	111.099	93.495	1.00	78.51	C
ATOM	6	CG	MET	A	1	105.176	110.881	94.812	1.00	76.25	C
ATOM	7	SD	MET	A	1	106.505	112.076	95.116	1.00	76.95	S
ATOM	8	CE	MET	A	1	107.077	111.551	96.763	1.00	73.31	C
ATOM	9	N	GLU	A	2	104.404	114.235	94.292	1.00	74.31	N
ATOM	10	CA	GLU	A	2	104.917	115.609	94.211	1.00	70.70	C
ATOM	11	C	GLU	A	2	105.963	115.909	93.143	1.00	68.10	C
ATOM	12	O	GLU	A	2	106.048	117.044	92.683	1.00	67.95	O
ATOM	13	CB	GLU	A	2	105.464	116.053	95.574	1.00	75.97	C
ATOM	14	CG	GLU	A	2	106.692	115.246	96.029	1.00	82.57	C
ATOM	15	CD	GLU	A	2	107.263	115.682	97.378	1.00	83.76	C
ATOM	16	OE1	GLU	A	2	106.611	115.396	98.412	1.00	86.90	O
ATOM	17	OE2	GLU	A	2	108.373	116.276	97.401	1.00	81.23	O
ATOM	18	N	ASN	A	3	106.789	114.924	92.784	1.00	64.50	N
ATOM	19	CA	ASN	A	3	107.834	115.134	91.773	1.00	59.93	C
ATOM	20	C	ASN	A	3	107.381	114.767	90.360	1.00	55.41	C
ATOM	21	O	ASN	A	3	108.159	114.856	89.416	1.00	53.33	O
ATOM	22	CB	ASN	A	3	109.086	114.308	92.095	1.00	62.56	C
ATOM	23	CG	ASN	A	3	109.531	114.441	93.535	1.00	62.42	C
ATOM	24	OD1	ASN	A	3	109.045	113.724	94.408	1.00	63.06	O
ATOM	25	ND2	ASN	A	3	110.484	115.326	93.787	1.00	62.31	N

Lecture 181

Online Resources for Protein Visualization

Background

- Protein structures can be visualized by tracing the CA atoms
- CA Coordinates can be taken from PDB



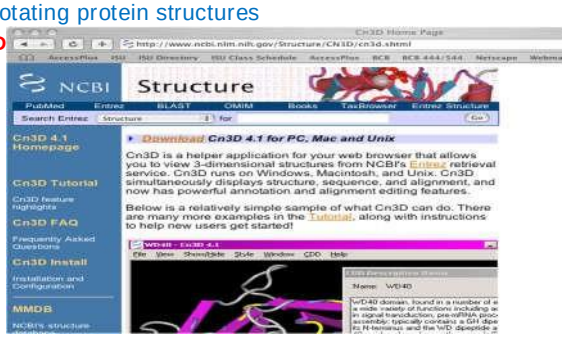
Online Tools

- Rasmol and CHIME are basic tools for visualizing proteins
- Swiss PDB Viewer offers several features such as protein surface view, alignment of several proteins & modelling secondary structures

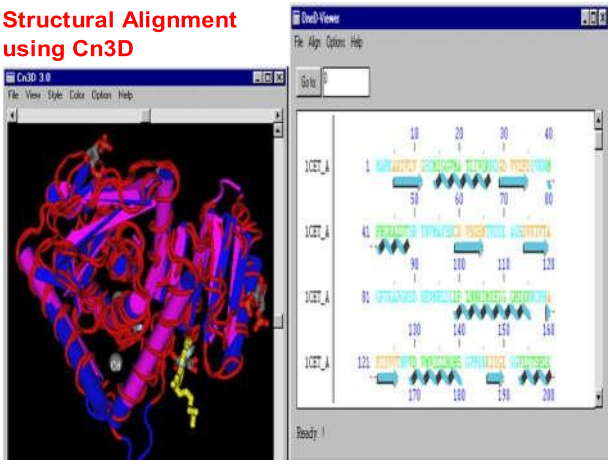
Online Tools

PyMOL is a python-script based tool for visualizing the protein structure. Cn3D is an other tool which helps us visualize protein structures. It also provides for annotating protein structures

Cn3D



Structural Alignment using Cn3D



Conclusion:

- Protein structures are visualized using several online tools
- These tools include Rasmol, CHIME, Swiss PDB Viewer and Cn3D.

Lecture 182

Types of Protein Visualizations

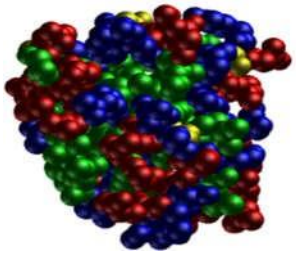
Background

- To visualize proteins, we use CA coordinates or positions
- We can use several online tools to view the resulting model
- What are the types of visualizations that can be generated by such online tools?

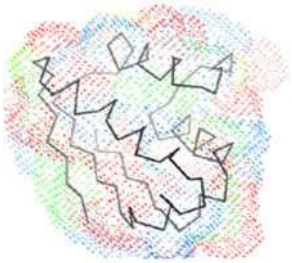
Introduction

- CPK: Corey-Paulin-Koltun Diagrams
- In CPK diagrams, each atom is represented by a solid sphere
- Spheres are equal to atomic van der Waal radius (the volume of the atom)

Sphere Diagram



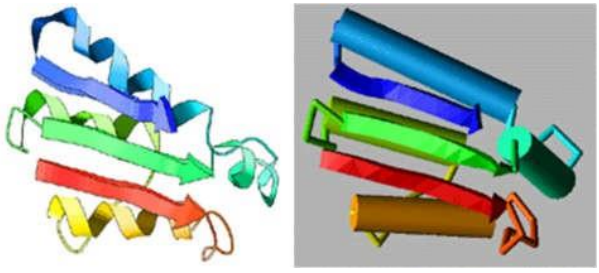
Surface Diagram



Ribbon Diagrams

- Ribbon diagrams are an easy and frequently used technique for representing protein structures
- Structure is represented by the secondary structures (fold) using simple cartoon figures.

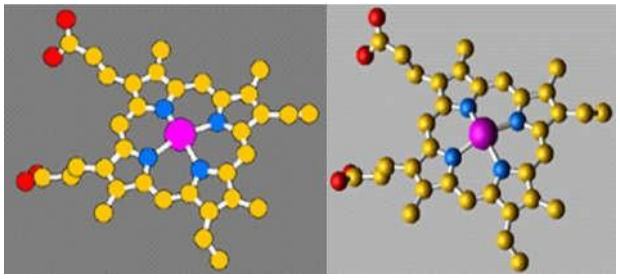
Ribbon Diagrams



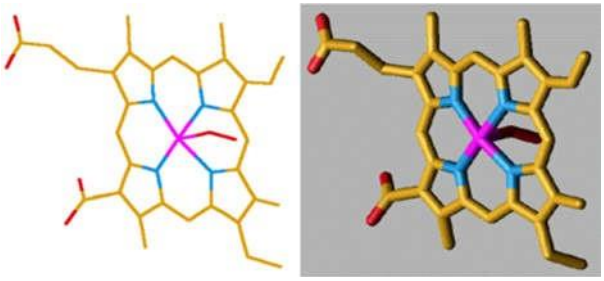
Balls & Stick (BS) Models

- BS model is another popular protein structure representation strategy
- BS Models have atoms as colored balls and intermediate bonds as sticks

Balls and Sticks Models



Colored Sticks Models



Conclusion:

- Protein Structure Visualization can be performed using several atomic representations
- These include CPK, Ribbon and Balls & Stick Diagrams

Lecture 183

Introduction to Energy of Protein Structures

Background:

- Proteins come together as a result of peptide bond formation between various amino acids
- The resulting polymer then goes through the step of folding which leads to the formation of a 3D structure

Introduction: Why do proteins fold?

Advantage of a folded protein over a denatured protein?

What are the factors that participate in folding the proteins?

Role of Amino Acids

We know that amino acids can be polar, charged and hydrophobic

Role of polar and charged amino acids in folding

Role of hydrophobic amino acids in folding

Overall Goal of Folding

- Anfinsen's thermodynamic hypothesis: Proteins fold for a unique, stable and minimum free kinetic energy structure
- What other factors may come into play for satisfying Anfinsen hypothesis

Minimizing Energy

- We know that if bonds can be formed between two atoms, then energy is released
- This leads to a situation where there is lesser free energy accessible to each atom for further interactions

Minimizing Energy

- So, proteins maximize bonds that can be made between the side chains on each of their constituent amino acids
- Which bonds can be formed in such a case?

Energies of Interactions

Force	Strength (kJ/mol)	Distance (nm)
Van der Waals	0.4-4.0	0.3-0.6
Hydrogen Bonds	12-30	0.3
Ionic Interactions	20	0.25
Hydrophobic Interactions	<40	varies

$$Energy_{TOTAL} = Atoms_{VWF} \times Energy_{VWF} + Atoms_{HB} \times Energy_{HB} + Atoms_{IonicInteraction} \times Energy_{IonicInteraction}$$

Conclusion

- Energies of a protein structures can be computed by 1st enumerating type of interactions b/w each atom.
- Then, accumulating the energy of each interaction towards calculating an overall energy of a protein.

Minimizing Energy

Such atomic interactions include:

- Disulphide bonds between Cysteine residues
- Hydrogen Bonds
- Van der Waals Forces
- Electrostatic Interactions between polar/charged amino acids

Conclusion

- The greater the number of these bonds, the more stable a protein becomes.
- Hence, the basic idea of thermodynamic stability is to maximize bonding in order to minimize the free energy

Lecture 185

Structure Determination for Energy Calculations

Background

- The greater the interactions between the amino acids, the more stable a protein becomes.
- We can calculate energy of a folded protein based on the number and types of atomic interactions.

How to find the number of interactions

- To determine the number of each type of interaction within a protein, we need to find its inter-atomic distances.
- Based on specific atomic distances, we can guess the type of atomic interaction
- By looking up at the bond/energy table, we can compute the overall energy.

Lecture 184

Calculating Energy of a Protein Structures

- The greater the number of bonds b/w the amino acid, the more stable a protein becomes.
- So how can we go about calculating energy of a folded protein?

Comparison of bond energies

- Hydrophobic interactions>
- Electrostatic interactions>
- Hydrogen bonds
- van der Waals

Calculating overall energy of a protein structure

- Given the No. of atomic interactions in a protein, you can simply sum the energy in the protein molecule!!

Energies of Interactions

Force	Strength (kJ/mol)	Distance (nm)
Van der Waals	0.4-4.0	0.3-0.6
Hydrogen Bonds	12-30	0.3
Ionic Interactions	20	0.25
Hydrophobic Interactions	<40	varies

$$Energy_{TOTAL} = Atoms_{VWF} \times Energy_{VWF} + Atoms_{HB} \times Energy_{HB} + Atoms_{IonicInteraction} \times Energy_{IonicInteraction}$$

Techniques for structure determination

- X-Ray Crystallography
- Nuclear Magnetic Resonance (NMR) Spectroscopy

Conclusion: We need to know the structure of the protein to calculate atomic distances. Atomic distances tell us about atomic interactions with neighboring atoms. To determine the structure, we use X-Ray or NMR

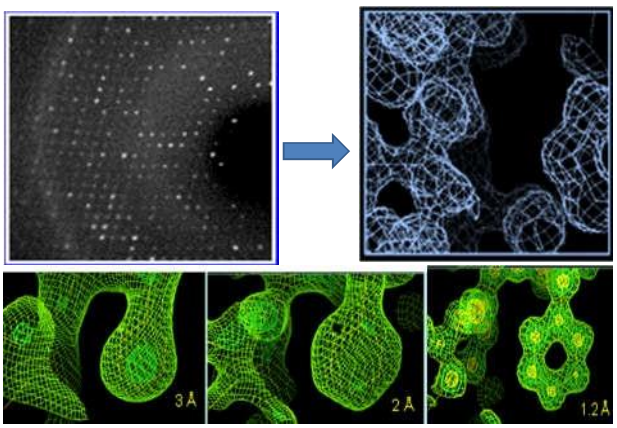
Lecture 186

Review of Experimental Structure Determination:

Background

- The greater the number of interactions b/w the amino acid side chains, the more stable protein becomes!
- These interactions define the final overall structure taken up by the protein.
- The structure also dictates which functions a protein can perform via the positioning of hydrophilic & polar amino acids.
- For determining stability, structure & function, we need to find the amino acid interactions.

From Diffraction Patterns to Atomic Positions



Introduction

- Several experimental methods exist for structure determination
- X-Ray Crystallography
- Nuclear Magnetic Resonance (NMR) Spectroscopy

To measure a bond/interaction, we must 1st see atoms!



Conclusion

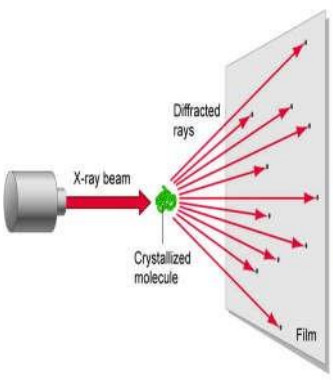
- Upon establishing the atomic positions and distances, we can then check for possible interaction b/w the different atoms.
- Atomic distances can help us classify interaction types e.g. hydrogen bonds, electrostatic & polar.

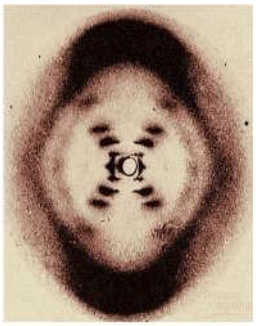
Lecture 187

Alpha Helices – I:

- Atomic distances can tell us about their existential interactions
- Different types of interactions may occur b/w atoms
- E.g. Hydrogen Bonds, Polar etc.]
- If two atoms are participating in a covalent bond, their distance is ~0.96A
- Incase of hydrogen bond formation b/w atoms, the inter-atomic distance is ~1.97A. X-Ray data should have a minimum of 1.97A resolution.

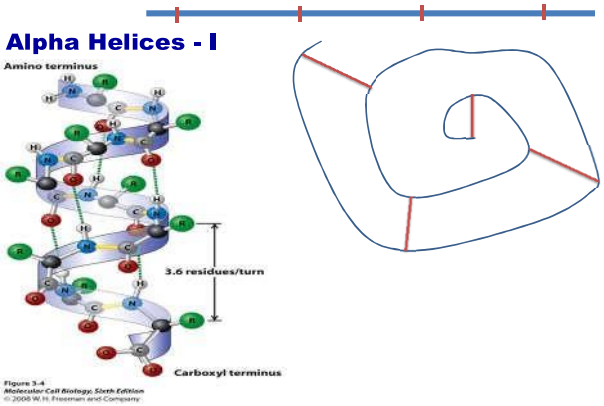
Principle of X-Ray Crystallography





Rosalyn Franklin's Diffraction pattern for DNA

Hydrogen Bonds to Fold an Amino Acid Chain



Conclusion

- X-Ray Crystallography data shows that Hydrogen atoms of N-Term may come together with Oxygen atoms of C-term amino acid at 4th neighboring position
- Their atomic distance is ~1.9A and hence are considered to be in a hydrogen bonds.

Lecture 188

Alpha Helices – II:

Background

- X-Ray Crystallography of protein shows that Hydrogen atoms of N-Term come together with Oxygen atoms of C-term amino acid at 4th neighboring position to make Hydrogen bonds!

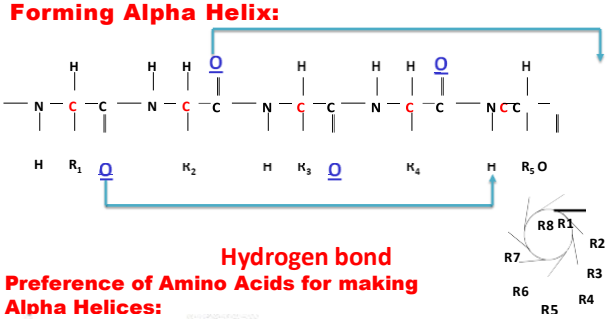
Helix Formers:

- From 20 amino acids, any one can be present in the backbone
- Is there a variable preference in amino acids to form helix?
- Yes,"Helix Formers" are generally hydrophobic amino acids (M, A, L…).

Conclusion:

- Alpha Helices are formed by hydrogen bonding (O-H) between C_i and N_{i+4} atoms in the protein backbone.
- What other types of secondary structures can be formed by Hydrogen bonding with in the protein backbone?

Forming Alpha Helix:



Preference of Amino Acids for making Alpha Helices:

Amino Acid	Preference: Helix	Preference: Strand	Preference: Turn	Properties
Glu	1.59	0.52	1.01	Helical preference; extended flexible side chains
Ala	1.41	0.72	0.82	
Leu	1.34	1.22	0.57	
Met	1.30	1.14	0.52	
Gln	1.27	0.98	0.84	
Lys	1.23	0.69	1.07	
Arg	1.21	0.84	0.90	
His	1.05	0.80	0.81	

Lecture 189

Beta Sheets - I

Background

- Alpha Helices are formed by hydrogen bonding (O-H) between C and N atoms in the protein backbone.

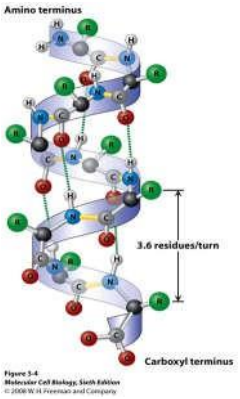
i i+4

- Is it possible that other types of secondary structures can also be formed?

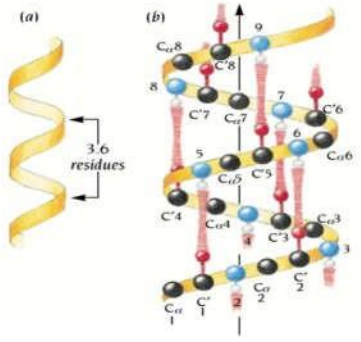
Introduction

- Beta Sheets are an other common secondary structure
- They are constituted by several Beta Strands which come together
- 5 to 10 resides are needed to make a Beta Strand, typically

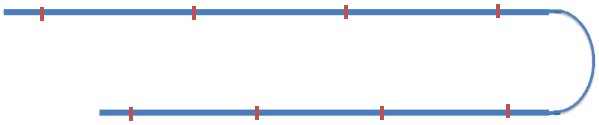
Every Oxygen bound to 4th neighboring Amino Group' Hydrogen



Carbons (Black) & Nitrogens (Blue): 1-5, 2-6, 3-7…

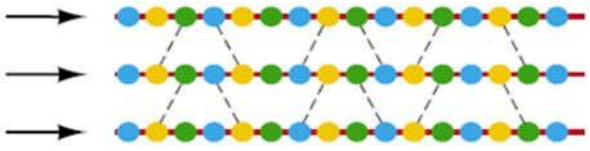


Hydrogen Bonds to make in Beta Strands



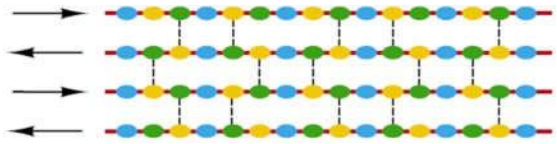
The Beta Sheet is made up of several Beta Strands

C-Alpha atoms and the CO and NH groups are shown in blue, yellow, and green, respectively.



This is called a parallel beta sheet!

C-Alpha atoms and the CO and NH groups are shown in blue, yellow, and green, respectively.



This is called an anti-parallel beta sheet!

Conclusion:

- Beta Sheets are an other secondary structure that can be formed as a result of hydrogen bonding between the protein back bone
- Do some amino acids have a preference for making Beta Sheets & what are the other types of Beta Sheets?

Lecture 190

Beta Sheets - II

Background

- Beta strands can make hydrogen bonds with each other and organize as beta sheets
- What are the other possible structures which beta strands can give rise to?

Beta Sheets Properties

1. Beta Strand
2. Beta Sheet
3. Beta Barrel
4. Beta Sandwiches

Beta Barrels

- Beta Barrel is made of a single beta sheet that twists and coils upon itself.
- The first strand in the beta sheet makes a hydrogen bonds with the last strand.

Beta Sandwiches

- Beta Sandwiches are made of 2 beta sheets which are usually twisted & packed so their strands are aligned

Preference of Amino Acids for making Beta Strands

Amino Acid	Preference			Properties
	Helix	Strand	Turn	
Val	0.90	1.87	0.41	Strand preference; bulky side chains, beta-branched
Ile	1.09	1.67	0.47	
Tyr	0.74	1.45	0.76	
Cys	0.66	1.40	0.54	
Trp	1.02	1.35	0.65	
Phe	1.16	1.33	0.59	
Thr	0.76	1.17	0.90	

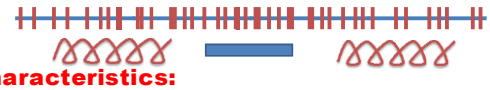
Conclusion: Beta Sheets are formed by H bonds b/w of 5–10 consecutive amino acids in one portion of the backbone with another 5–10 farther down the backbone. Beta strands may be adjacent (with a loop in b/w) or far with other structures in between.

Lecture 191

Loops – I:

- Alpha Helices and Beta Sheets are secondary structures formed as a result of hydrogen bonding in between protein backbone atoms.
- How are Alpha Helices and Beta Sheets joined?

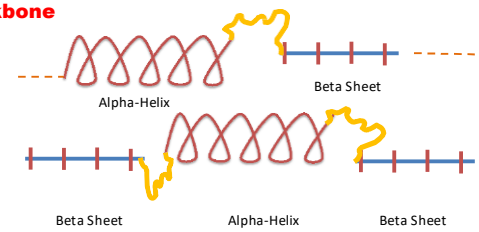
Protein Backbone & Secondary Structures



Characteristics:

- Loops are mostly located on the surface of protein structure
- Mutate in sequence at a much faster rate than Alpha Helices & Beta Sheets
- Loops are flexible & can adopt multiple conformations

Joining Alpha Helices & Beta Sheets in a Protein Backbone



Loops are formed by amino acids present in the middle of the Alpha Helices and Beta Sheets in a protein backbone.

Loops are variable in length and 3-D conformations



Variability in length & conformation allows loops to join Alpha Helices & Beta Sheets in a variety of ways!

Conclusion:

- Loops dictate the overall structure of protein as they couple Alpha helices and beta sheets
- Do they have a role in protein function as well?
- Which amino acids prefer to form loops?

Lecture 192

Loops - II

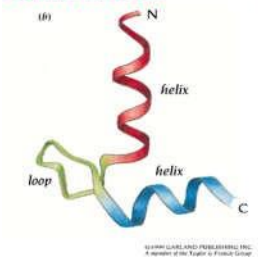
Background

- Loops dictate the overall structure of protein as they couple Alpha helices and Beta sheets
- Loops are flexible and have variable lengths so as to successfully bridge between secondary structures

Loop Properties:

- Loops are mostly comprised of charged & polar amino acids
- Loops frequently participate as components of active site

Loops in 3D Conformation



Preference of Amino Acids for making Loops

Table 9.2. *Chemical properties of the 20 amino acids*

Chemical group	Amino acid (one-letter code)	Name
Charged		
	D	aspartic acid
	E	glutamic acid
	K	lysine
	R	arginine
Polar		
	S	serine
	T	threonine
	Y	tyrosine
	H	histidine
	C	cysteine
	N	asparagine
	Q	glutamine
	W	tryptophan

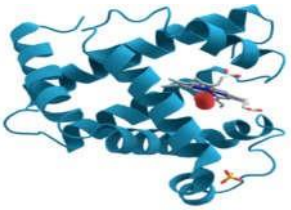
Types of Loops

- Hairpin loops are two amino acids long & join anti-parallel Beta strands
- Other Loops may be 3 to 4 amino acids long
- Loops fall into various families

Conclusion:

- Loops are the third type of secondary structure after Alpha helices and Beta sheets
- Loops are unique in that they are flexible & variable length
- Loops constitute active sites!!

Coils in Myoglobin



Conclusion:

- Coils are those secondary structure formed by the protein backbone which are neither helices, sheets or loops
- In fact, coils do not have a consistent classifiable structure
- Hence, coils are random structure and random length

Lecture 193

Coils:

- Alpha helices & beta sheets are the regular secondary structures
- Loops are flexible secondary structures & connect alpha helices & beta sheets
- Is there any other type of secondary structure?
- Coils are an other secondary structure
- Coils are unstructured and unlike loops
- Essentially, a secondary structure which is not a helix, sheet or loop is a coil!

Functional Aspects of Coils

- Coils are apparently disordered regions
- They are oriented randomly while being bonded to adjacent amino acids. However, coils also appear to play important functional roles.

Lecture 194

Structure Classification - I

Background

- Proteins have primary, secondary, tertiary & quaternary structures.
- Each level of protein structure organization is known to impart specific characteristics to the protein.

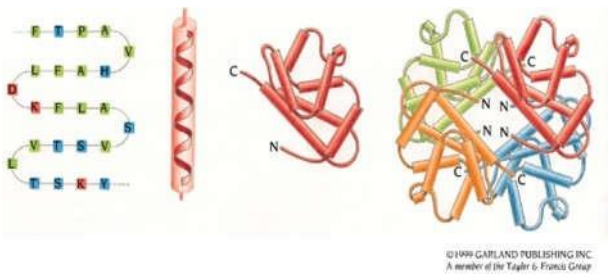
Review of the 4 structure levels

1. Primary Structures
2. Secondary Structures
3. Tertiary Structures
3. Quaternary Structures

Introduction

- Structural artifacts tend to be more conserved as compared to their sequences
- Therefore, it may be useful to look at the secondary/tertiary structures for conservation study

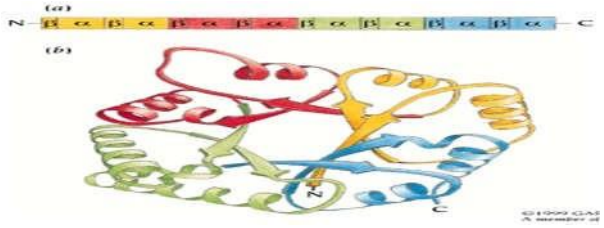
Primary Secondary Tertiary Quaternary



Classification:

- The evolution of protein structures and their hierarchy is not systematized
- Hence, we need to classify the function of protein by examining their secondary & tertiary structures.

Motifs (Non-functional Combinations of 2' structures)



Domain (Functionally Complete)

Conclusion:

- Domains are semi-independent functional structures in a protein
- Have a stable structure
- Over ~40 residues
- Protein may contain multiple domains

Lecture 195

Structure Classification - II

- Domains are semi-independent functional structures in a protein
- Protein may contain multiple domains
- Hence, we can try to classify proteins by their domains

Identifying Domains?

- Locally Compact – Domains interact (H-bonds) more internally than externally
- Domains have a hydrophobic core
- Domains are contiguous (min. chain breaks

- Domains have a minimal contact with rest of the peptide
- Solvent area in contact with each domain should not vary significantly upon separating 2 separate domains

Types of Domains

- Alpha Domains
- Beta Domains
- Alpha/Beta Domains
- Alpha + Beta Domains
- Alpha & Beta Multi-Domains
- Membrane & cell-surface proteins

Conclusion:

- So, by looking at proteins, we can list the domains present in each protein
- Once domains in each protein are listed, we can classify whole proteins into various types and classes!

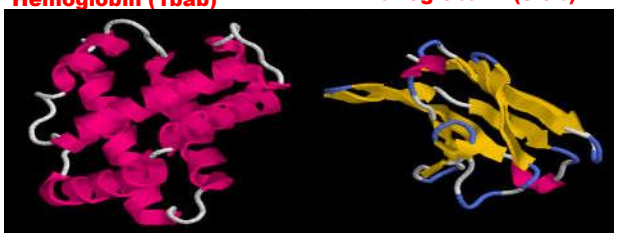
Lecture 196

Examples of Protein Domains

1. Alpha Domains
2. Beta Domains
3. Alpha/Beta Domains
4. Alpha + Beta Domains
5. Alpha & Beta Multi-Domains
6. Membrane & cell-surface proteins

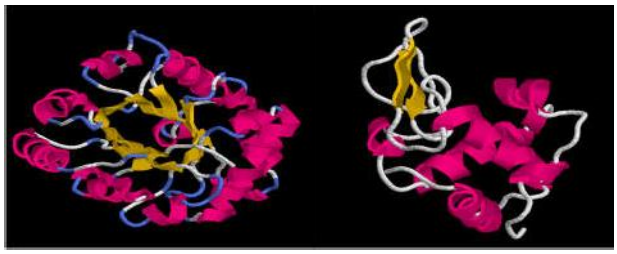
Alpha Domain: Hemoglobin (1bab)

Beta Domain: Immunoglobulin (8fab)



Alpha / Beta: Triosephosphate isomerase (1hti)

Alpha + Beta: Lysozyme (1jsf)



Conclusion:

- Various types of domain architectures exist in proteins
- Such architectures can be classified into general structural classes
- Databases can be made from classes.

Lecture 197

CATH Classification

Background

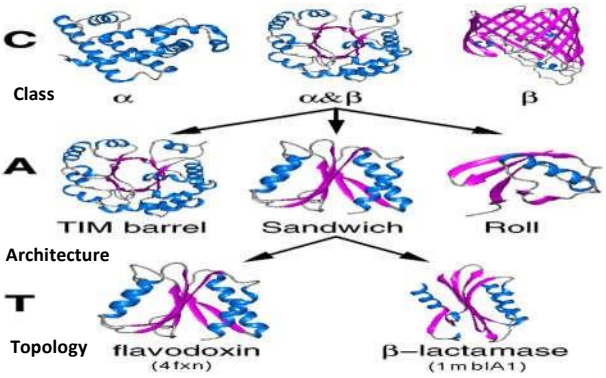
- Domains can be classified into structural classes
- Classes can be further classified into Architecture & Topologies
- Let's see how it is done in CATH.

Class

- Similar secondary structure content
- All α, all β, alternating α/β etc

Architecture

- Also called FOLD
- Major structural similarity
- SSE's in similar arrangement



www.biochem.ucl.ac.uk/bsm/cath_new/index.html

Lecture 198

Classification Databases

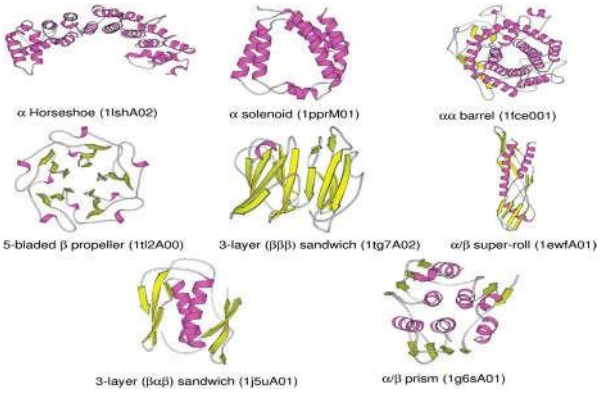
Background

- Proteins are classified into various structural classes
- CATH is one such system in which proteins are organized into classes, architecture, topology & homology.

Conclusion:

- FSSP - Family of Structurally Similar Proteins, based on the DALI algorithm
- Pclass - Protein Classification, based on the LOCK and 3Dsearch algorithms

Structural Classes:





Topology

- Super Family
- Probable common ancestry
- Family membership

Homology

- Same Family
- Clear evolutionary relationship
- Pairwise sequence similarity > 30%

Conclusion

- CATH classifies proteins by their structural similarity
- It also considers the internal organization of the structural components in proteins
- Databases using CATH classification?



SCOP Classification Statistics

SCOP: Structural Classification of Proteins. 1.75 release
38221 PDB Entries (23 Feb 2009). 110800 Domains. 1 Literature Reference
(excluding nucleic acids and theoretical models)

Class	Number of folds (Architecture)	Number of superfamilies (Topology)	Number of families (Homology)
All alpha proteins	284	507	871
All beta proteins	174	354	742
Alpha and beta proteins (a/b)	147	244	803
Alpha and beta proteins (a+b)	376	552	1055
Multi-domain proteins	66	66	89
Membrane and cell surface proteins	58	110	123
Small proteins	90	129	219
Total	1195	1962	3902

http://scop.mrc-lmb.cam.ac.uk/scop/count.html

Lecture 199

Algorithms for Structure Classification

1. CATH – Class, Architecture, Topology and Homology
2. SCOP – Structural Classification of Proteins
3. FSSP - Family of Structurally Similar Proteins
4. Pclass – Protein Class

- So how exactly do we compare protein domains or structure with each other to conclude their relationship?
- Several algorithms exist for classifying protein structures

Intra-Molecular Distance Algorithms

- Proteins are considered as rigid bodies
- They are placed in a 3D Cartesian coordinate system
- Structural alignment in 3D E.g. VAST, LOCK

Inter-Molecular Distance Algorithms:

- Proteins are considered as rigid bodies
- They are placed in 2D
- Structural alignment using internal distances & angles

- The basic idea is to capture internal geometry of protein structures
- E.g. DALI, and SSAP

Conclusion:

- Such algorithms are also very useful to compare whole protein structures
- They can help determine evolutionary relationship
- Also, functional similarity can be estimated!

Lecture 200

Protein Structure Comparison

Introduction

- Proteins are assembled into primary (1'), secondary (2'), tertiary (3') and quaternary (4') structures
- Protein sequence is less conserved than its structure
- Protein structure determines function.

Why Compare two Protein Structures

- Since protein structure dictates function, comparing two structures can help us evaluate if the proteins do the same or similar function
- Comparing whole vs partial structure?

Comparing Whole Protein Structures

- Proteins contain multiple structural subunits e.g. secondary structures, motifs & domains
- Structures of all such subunits are to be considered as one and compared!

Comparing Portions within a Protein Structure

- We know that domains are functionally independent components of the protein structure
- Proteins may have multiple domains

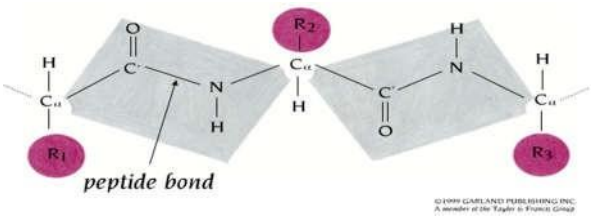
Comparing Portions within a Protein Structure

- So for two different proteins, sharing the same domain, we may want to compare only a portion of the overall structure i.e. a domain.

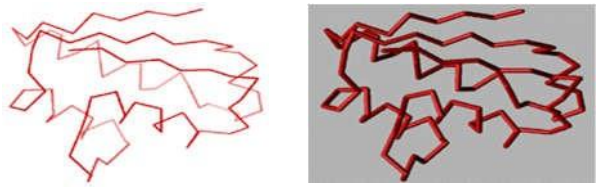
What to Compare?

- For comparing the complete or partial protein structures, the position of Alpha Carbon atoms can be used.
- The (x,y,z) positions of Alpha Carbon atoms can be obtained from the PDB.

C-Alpha atoms in backbone



Tracing and Visualizing C-Alpha Backbone



http://www.danforthcenter.org/smith/MolView/Over/overview.html

Conclusion

- PDB coordinates of Alpha Carbons in the protein back bone can be used for comparison.
- In this way, whole protein structure or domains etc. can be compared.

Lecture 201

Strategies for Structure Comparison - I

Introduction

- PDB coordinates of Alpha Carbons in the protein back bone can be used for comparison.
- Thus, two whole protein structures or domains within each structure can be compared.

Tracing and Visualizing C-Alpha Backbone

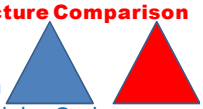


http://www.danforthcenter.org/smith/MolView/Over/overview.html

Strategy # 1 – Whole Protein Structure Comparison by Intermolecular distances

1. Two protein sequences are pair-wise aligned with each other
2. Corresponding Alpha Carbons are identified
3. Coordinates of corresponding Alpha Carbons are retrieved from PDB
4. Their individual differences calculated
5. Root Mean Square Distance is computed to assess the similarity

Whole Structure Comparison



Conclusion:

- Whole protein structures can be compared by calculating the root mean squared difference (RMSD) b/w their Alpha Carbons positions
- The lower the RMSD, the similar are the proteins.

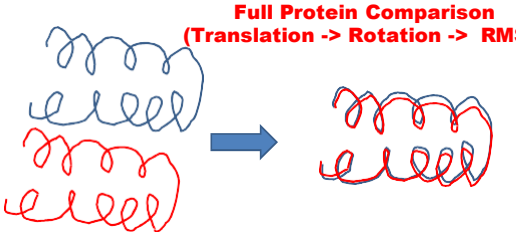
Lecture 202

Strategies for Structure Comparison - II

Introduction

- Full protein structures can be compared & ranked by the overall differences in positions b/w their Alpha Carbons.
- But proteins are 3D and in various conformations

Full Protein Comparison (Translation -> Rotation -> RMSD)



Calculating RMSD – An Example

$$RMS(A,B) = \sqrt{\frac{1}{N}\sum_{i=1}^{N} d(a_i,b_i)^2}$$

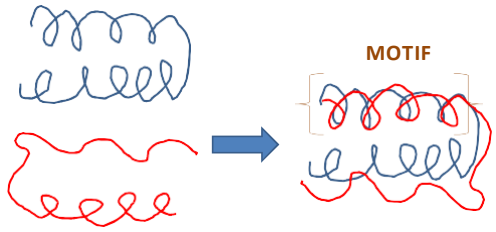
where $d\ (a_i - b_i)^2 = (x_2 - x_1)^2 + (y_2 - y_1)^2 + (z_2 - z_1)^2$



Conclusion

- Motifs, Domains and Full Proteins can be compared by using the rigid body super-positioning
- Depending on the RMSD, proteins, their motifs and domains can be selectively compared.

Domain or Motif Comparison (Region Selection ->Translation -> Rotation -> RMSD)



Lecture 203

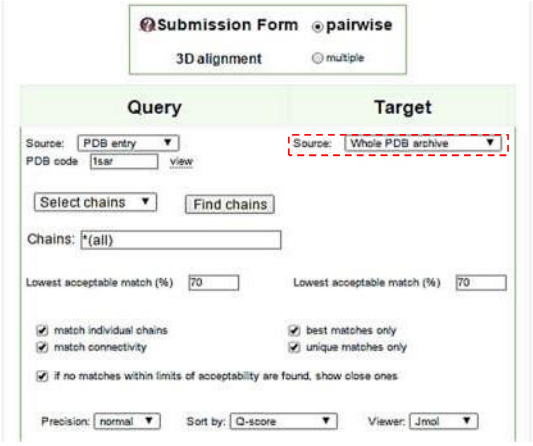
Online Resources for Structure Comparison

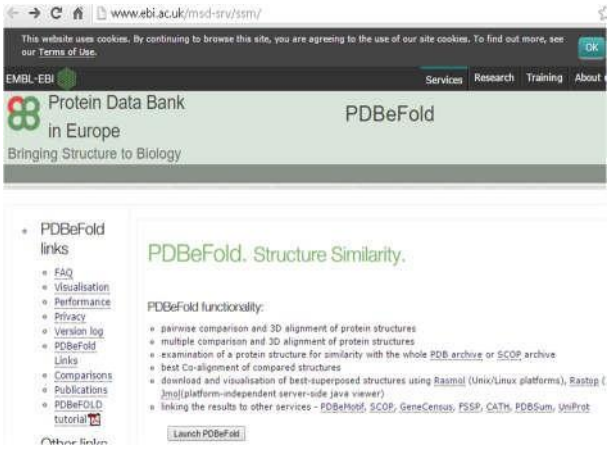
Introduction

- Multiple types of comparison can be performed b/w Proteins, Motifs, & Domains by rigid body super-positioning
- RMSD tells us about the quality of the matches

Conclusion

- Protein structures can be compared in multiple ways
- Till now, we can compare proteins by their motifs, domains and full structures
- There are several advanced techniques for this as well.





Lecture 204

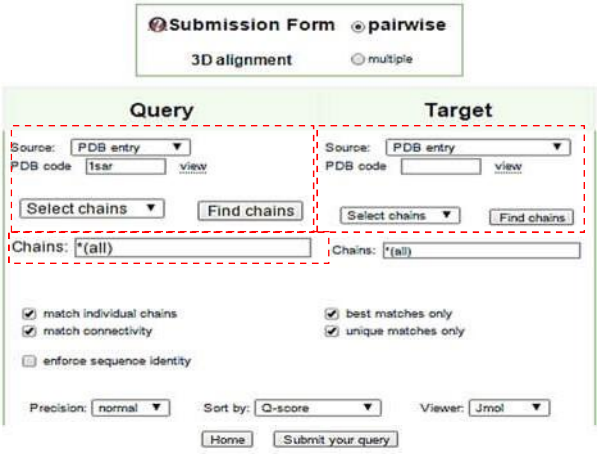
Protein Structure Prediction

Background

- Complex protein structures enable proteins to perform complex functions
- We know over a million protein sequences but only about 100,000 protein structures. Why?
- Estimating exact protein structures is very difficult
- Its difficult to crystallize proteins
- Even if we manage to get protein's X-Ray, to reconstruct the structure is extremely complex

Introduction

- What if we could somehow predict protein structures?
- Since we know so many sequences, they can be used for predicting protein structures
- This indeed is possible and helpful



The Basic Idea

1. Amino acids determine the protein structure
2. We have a large protein sequence dataset (uniprot)

Hence, we can fold protein sequences and predict their structures!

Why predict and why not exact solutions?

A deterministic solution of protein folding is a major unsolved problem in molecular biology!

Proteins fold spontaneously or with the help of enzymes or chaperones.

To predict we must first learn!

To computationally predict protein structures, we need to copy or mimic the natural folding! What are the steps in protein folding & structure formation?

To fold we must learn the steps

Step 1: "Collapse"- leading to burial of hydrophobic AA's
Step 2: Fluid globule - helices & sheets form, but are unorganized
Step 3: Compaction, and rearrangement of 2' structures

Conclusion

- Protein structure prediction involves learning how the amino acids in primary sequence fold.
- Using this information, upon getting a protein sequence, we can try to predict how it folds!

Lecture 205

Predicting Secondary Structures

- Since the first step in protein folding is the formation of secondary structures, we must evaluate which amino acids in the primary sequence prefer which secondary structures?
- By looking at the structures in PDB, we know that Alanine mostly found in Alpha Helices
- So if we have several Alanines in the sequence, then

  we can anticipate that a helix may be formed by them
- What if we survey the entire PDB and check the presence of each amino in each type of secondary structure
- If we know which amino acid is found in which specific secondary structure, then we can use it for prediction!

Chou & Fasman (1974 & 1978)

Amino Acid	P_α	P_β	P_t
Glu	1.51	0.37	0.74
Met	1.45	1.05	0.60
Ala	1.42	0.83	0.66
Val	1.06	1.70	0.50
Ile	1.08	1.60	0.50
Tyr	0.69	1.47	1.14
Pro	0.57	0.55	1.52
Gly	0.57	0.75	1.56

Conclusion:

- Several algorithms have been designed to predict 2' given an amino acid sequence
- The first such algorithm was the Chou-Fasman Algorithm!
- We will see it in the upcoming modules!

Lecture 206

Introduction to Chou Fasman Algorithm

Background

- 3D Structure of proteins is determined by their Amino Acid sequence
- Note that we only know 100,000 3D protein structures, but 10 times more sequences!

Introduction

- For those proteins whose structure is already known, can we evaluate their amino acid sequence to find out which amino acids make which type of secondary (2') structures?

Predicting the 2' structures:

- Now, lets consider that if we are given an amino acid sequence, we can simply look up the propensity table and assign the tentative secondary structure

Sequence: E M A V I Y P G
2' Structure: α α α β β β T T

1.51 * 1.45 * 1.42 * 1.7 * 1.6 * 1.47 * 1.52 * 1.56

= 29.47

The propensity for forming "ααα β β β T T" is = 29.47

	P_α	P_β	P_t
Glu	1.51	0.37	0.74
Met	1.45	1.05	0.60
Ala	1.42	0.83	0.66
Val	1.06	1.70	0.50
Ile	1.08	1.60	0.50
Tyr	0.69	1.47	1.14
Pro	0.57	0.55	1.52
Gly	0.57	0.75	1.56

Chou & Fasman (1974 & 1978)

Conclusion:

- Given an amino acid sequence, look up the propensity table for each amino acid's propensity for various 2' structures
- Product of these propensity values will give you the overall propensity for formation of each 2' structure

Lecture 207

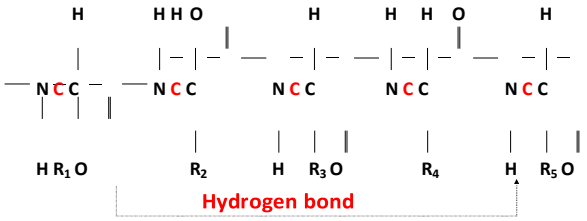
2' Structures in Chou Fasman Algorithm

- For a primary sequence, and a tentative 2' structure, propensity table can help us compute the overall propensity
- Product of propensity values is computed for overall propensity for each 2' structure.
- An important point to note here is that 2' structures are formed due to hydrogen bonding between amino acids
- So, we need to consider the neighboring amino acids

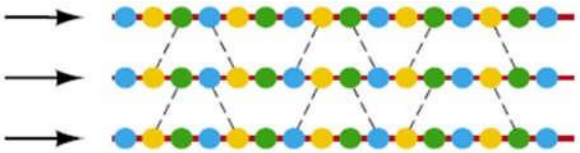
  as well!

Conclusion: You only need to compute propensities for a small number 2' structures. The highest net propensity will be the most probably secondary structure that will be formed!

Alpha Helix



Beta Sheet



Name	P(a)	P(b)	P(turn)	f(i)	f(i+1)	f(i+2)	f(i+3)
Alanine	1.42	0.83	0.66	0.06	0.076	0.035	0.058
Arginine	0.98	0.93	0.95	0.070	0.106	0.099	0.085
Aspartic Acid	1.01	0.54	1.46	0.147	0.110	0.179	0.081
Asparagine	0.67	0.89	1.56	0.161	0.083	0.191	0.091
Cysteine	0.70	1.19	1.19	0.149	0.050	0.117	0.128
Glutamic Acid	1.39	1.17	0.74	0.056	0.060	0.077	0.064
Glutamine	1.11	1.10	0.98	0.074	0.098	0.037	0.098
Glycine	0.57	0.75	1.56	0.102	0.085	0.190	0.152
Histidine	1.00	0.87	0.95	0.140	0.047	0.093	0.054
Isoleucine	1.08	1.60	0.47	0.043	0.034	0.013	0.056
Leucine	1.41	1.30	0.59	0.061	0.025	0.036	0.070
Lysine	1.14	0.74	1.01	0.055	0.115	0.072	0.095
Methionine	1.45	1.05	0.60	0.068	0.082	0.014	0.055
Phenylalanine	1.13	1.38	0.60	0.059	0.041	0.065	0.065
Proline	0.57	0.55	1.52	0.102	0.301	0.034	0.068
Serine	0.77	0.75	1.43	0.120	0.139	0.125	0.106
Threonine	0.83	1.19	0.96	0.086	0.108	0.065	0.079
Tryptophan	1.08	1.37	0.96	0.077	0.013	0.064	0.167
Tyrosine	0.69	1.47	1.14	0.082	0.065	0.114	0.125
Valine	1.06	1.70	0.50	0.062	0.048	0.028	0.053

Sequence: E M A V I Y P G

Viable 2' Structures:
α α α α β β β β β
α α α α α α α α β …
…α β β β β β α α α …

	P_α	P_β	P_t
Glu	1.51	0.37	0.74
Me	1.45	1.05	0.60
Ala	1.42	0.83	0.66
Val	1.06	1.70	0.50
Ile	1.08	1.60	0.50
Tyr	0.69	1.47	1.14
Pro	0.57	0.55	1.52
Gly	0.57	0.75	1.56

Chou & Fasman (1974 & 1978)

Very small number of 2' structural combinations are left!

1. Scan through the sequence : E M A V I Y P G
2. Identify sequence regions where:
   - 4 out of 6 contiguous residues give a P(α) > 1.0
   - That region is declared as alpha-helix
   - Extend helix to both sides until 4 out of 6 contiguous residues give a P(α) < 1.0
   - That is declared end of the helix

Conclusion:
- For Alpha Helices, 4 contiguous amino acids are required
- Their Alpha-Helix propensity should be more than 1.0
- Once this propensity falls below 1.0, Alpha-Helix stops

Lecture 208

Chou Fasman Algorithm - I

Background
- Only a small number of combinations of secondary structures are possible due to their individual properties
- Such as 4 amino acids are needed to start an Alpha Helix and 5 amino acids for Beta Sheet.

Introduction
- Note that besides the alpha helix and beta sheets, LOOPS are an other secondary structure
- How can loops be integrated into predicting 2' structures?
- Loops are small ~ 3-4 amino acids

Lecture 209

Chou Fasman Algorithm - II

Background
- Alpha Helices are formed from 4 contiguous amino acids having an Alpha-Helix propensity over 1.0
- The Alpha-Helix stops if this propensity falls below 1.0!

Introduction
- Once Alpha Helices are constructed, and concluded, the remaining amino acids can be evaluated for Beta sheets and turns etc
- Let's see how Beta sheets are evaluated using Chou Fasman Algorithm

Name	P(a)	P(b)	P(turn)	f(i)	f(i+1)	f(i+2)	f(i+3)
Alanine	1.42	0.83	0.66	0.06	0.076	0.035	0.058
Arginine	0.98	0.93	0.95	0.070	0.106	0.099	0.085
Aspartic Acid	1.01	0.54	1.46	0.147	0.110	0.179	0.081
Asparagine	0.67	0.89	1.56	0.161	0.083	0.191	0.091
Cysteine	0.70	1.19	1.19	0.149	0.050	0.117	0.128
Glutamic Acid	1.39	1.17	0.74	0.056	0.060	0.077	0.064
Glutamine	1.11	1.10	0.98	0.074	0.098	0.037	0.098
Glycine	0.57	0.75	1.56	0.102	0.085	0.190	0.152
Histidine	1.00	0.87	0.95	0.140	0.047	0.093	0.054
Isoleucine	1.08	1.60	0.47	0.043	0.034	0.013	0.056
Leucine	1.41	1.30	0.59	0.061	0.025	0.036	0.070
Lysine	1.14	0.74	1.01	0.055	0.115	0.072	0.095
Methionine	1.45	1.05	0.60	0.068	0.082	0.014	0.055
Phenylalanine	1.13	1.38	0.60	0.059	0.041	0.065	0.065
Proline	0.57	0.55	1.52	0.102	0.301	0.034	0.068
Serine	0.77	0.75	1.43	0.120	0.139	0.125	0.106
Threonine	0.83	1.19	0.96	0.086	0.108	0.065	0.079
Tryptophan	1.08	1.37	0.96	0.077	0.013	0.064	0.167
Tyrosine	0.69	1.47	1.14	0.082	0.065	0.114	0.125
Valine	1.06	1.70	0.50	0.062	0.048	0.028	0.053

Scan the sequence to identify regions where:

- 3 out of 5 amino acids have P(β) > 1.0
- That region is declared as beta sheet
- Extend beta sheet to both sides until 4 contiguous residues average P(β) < 1.0
- That is declared end of the beta sheet
- Those regions are finalized as beta-sheets which have average P(β) > 1.05 and the average P(β) > P(α) for that region.

Regions where overlapping alpha-helices and beta-sheets occur are declared helices if

- the average P(a-helix) > P(b-sheet) for that region

Else, a beta sheet is declared if

- average P(b-sheet) > P(a-helix) for that region.

1. Compute P(β) for contiguous regions of 5 Amino Acids
2. From these regions, identify regions where:
   - 5 contiguous residues have P(α) > P(β)
   - That region is finalized as alpha-helix

Repeat this step for the full amino acid sequence to finalize all possible alpha helical regions in the sequence.

Conclusion

- Alpha Helices can be finalized if their propensity is higher than the propensity for Beta Sheets in regions of 5 amino acids
- For those regions where that is not the case, further evaluation is required

Lecture 211

Chou Fasman Algorithm - IV

Background

- After computing the propensity of alpha helices and beta sheets, we need to settle for loops
- Let's see how can we find out the loops using Chou Fasman Algorithm.

Conclusion

- Chou Fasman Algorithm helps predict Alpha Helices, Beta Sheets and Turns
- The algorithm is based on statistical occurrence of Amino Acids in known structures.

Lecture 210

Chou Fasman Algorithm - III

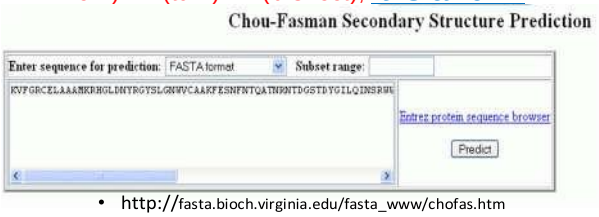
- Alpha Helices were finalized if their propensity was higher than the propensity for Beta Sheets in regions of 5 amino acids.
- For those regions where that is not the case, what should be done?
- We can evaluate such regions for Beta Sheets
- Let us see step by stop how to find a beta sheet and how to differentiate them from alpha helices.

Conclusion:

- Using the strategy of higher propensity, alpha helices and beta sheets can be completely resolved
- Assignments for each beta sheet and alpha helix can be finalized. But what about the loops?

Name	P(a)	P(b)	P(turn)	f(i)	f(i+1)	f(i+2)	f(i+3)
Alanine	1.42	0.83	0.66	0.06	0.076	0.035	0.058
Arginine	0.98	0.93	0.95	0.070	0.106	0.099	0.085
Aspartic Acid	1.01	0.54	1.46	0.147	0.110	0.179	0.081
Asparagine	0.67	0.89	1.56	0.161	0.083	0.191	0.091
Cysteine	0.70	1.19	1.19	0.149	0.050	0.117	0.128
Glutamic Acid	1.39	1.17	0.74	0.056	0.060	0.077	0.064
Glutamine	1.11	1.10	0.98	0.074	0.098	0.037	0.098
Glycine	0.57	0.75	1.56	0.102	0.085	0.190	0.152
Histidine	1.00	0.87	0.95	0.140	0.047	0.093	0.054
Isoleucine	1.08	1.60	0.47	0.043	0.034	0.013	0.056
Leucine	1.41	1.30	0.59	0.061	0.025	0.036	0.070
Lysine	1.14	0.74	1.01	0.055	0.115	0.072	0.095
Methionine	1.45	1.05	0.60	0.068	0.082	0.014	0.055
Phenylalanine	1.13	1.38	0.60	0.059	0.041	0.065	0.065
Proline	0.57	0.55	1.52	0.102	0.301	0.034	0.068
Serine	0.77	0.75	1.43	0.120	0.139	0.125	0.106
Threonine	0.83	1.19	0.96	0.086	0.108	0.065	0.079
Tryptophan	1.08	1.37	0.96	0.077	0.013	0.064	0.167
Tyrosine	0.69	1.47	1.14	0.082	0.065	0.114	0.125
Valine	1.06	1.70	0.50	0.062	0.048	0.028	0.053

- For any *jth* residue in sequence, we calculate f (Total) = f(j) f(j+1) f(j+2) f(j+3) (tetrapeptide)
- If
  1. f(Total) > 0.000075
  2. the average value for P(turn) > 1.00 in the tetrapeptide
  3. the averages for the tetrapeptide are such P(a-helix) < P(turn) > P(b-sheet), It is a Turn!



- http://fasta.bioch.virginia.edu/fasta_www/chofas.htm

Lecture 212

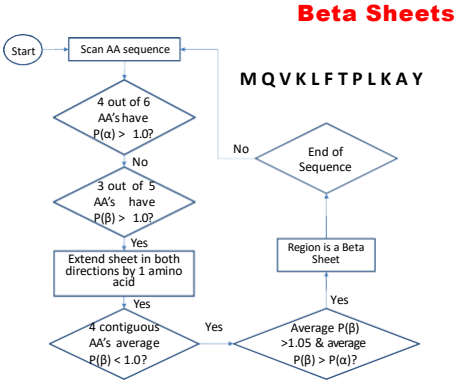
Chou Fasman Algorithm – Flowchart I

Background

- Chou Fasman Algorithm helps predict secondary structures such as Alpha Helices, Beta Sheets & Turns
- Step by step flowchart of the entire algorithm

Conclusion

- Beta sheets can be predicted from primary amino acid sequences
- Next, we will see the flowchart of Alpha Helices and Beta Turns



Lecture 213

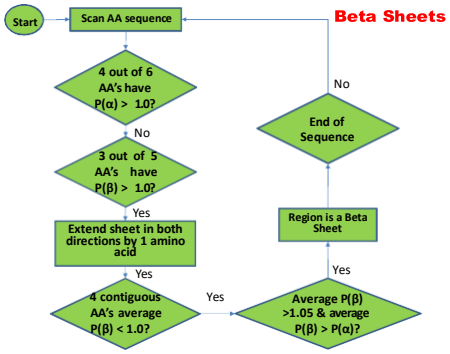
Chou Fasman Algorithm – Flowchart II

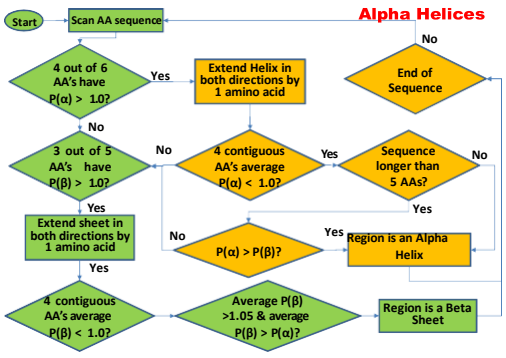
Background

- Chou Fasman Algorithm helps predict secondary structures from amino acid sequences
- Step by step flowchart of the algorithm for extracting Alpha Helices

Conclusion

- Now we have reviewed flowcharts for Alpha Helices and Beta Sheets
- Next up is the flow chart for Beta Turns.





Lecture 214

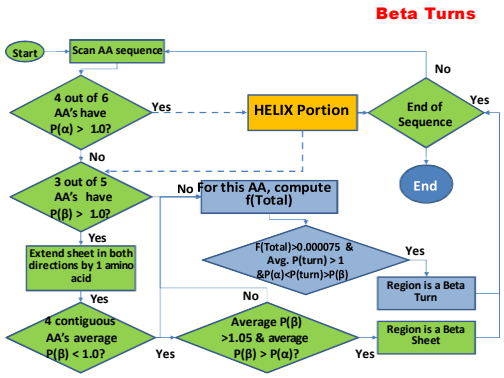
Chou Fasman Algorithm – Flowchart III

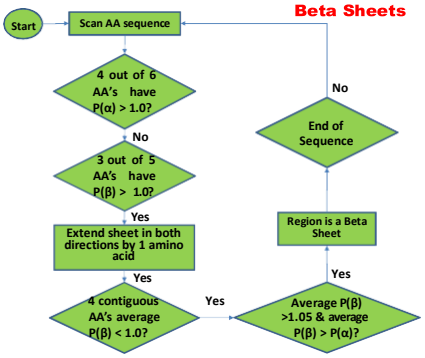
Background

- Chou Fasman Algorithm helps predict secondary structures such as Alpha Helices, Beta Sheets & Turns
- Step by step flowchart of the entire algorithm

Conclusion

- Alpha helices, beta sheets and turns can be predicted using Chou Fasman Algorithm
- This algorithm is based on statistical analysis of amino acid occurrences in proteins.





Lecture 215

Chou Fasman Algorithm – Improvements

Background

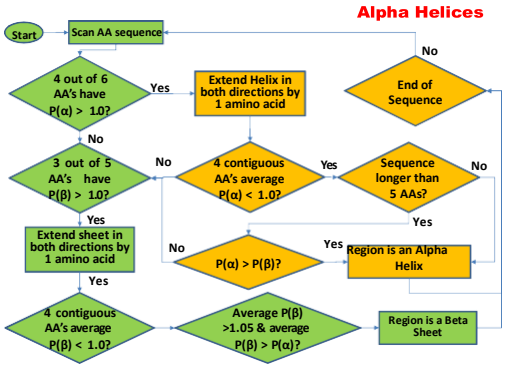
- Alpha helices, beta sheets and turns can be predicted using Chou-Fasman Algorithm
- The algorithm is based on statistical analysis of amino acid occurrences in proteins

Improvements

- Secondary structure propensity values of alpha helix, beta sheet and turns should be recalculated with the latest protein data sets.

Special consideration for:

- Nucleation regions
- Membrane proteins
- Hydrophobic domains



Consider variable coil and loop sizes besides the from tetra peptide turns

- Consider local protein folding environments
- Solvent accessibility of residues
- Protein structural class
- Protein's organism

Conclusion:

- Chou Fasman can be improved to better predict secondary structures by incorporating biochemical factors and updated statistics!

Lecture 216

Summary of Visualization, Classification & Prediction

Structure Visualization

1. Why do we need to visualize proteins?
2. Which atoms are used to reconstruct proteins?
3. Where are the positions of these atoms stored?

Structure Classification

1. What is the relationship b/w protein structure & function?
2. What is the need to classify proteins
3. Hierarchy of classification

Structure Prediction

1. Why structure of proteins are important?
2. Why are so few structures reported till date?
3. Benefits of predicting structures

Conclusion:

Structure visualization, classification and prediction equip us to perform functional evaluation of proteins! This is important for understanding disease and designing drugs for treating them!

Lecture 217

Introduction to Homology Modelling

- Proteins are 3D molecules with their own unique structures
- Protein structure is reflective of the protein function
- Protein structure includes 1', 2', 3' and 4' structures
- 1'structure of proteins is the sequence of proteins and can be obtained by mass spectrometry
- 2'structures formed by proteins are the helices, beta sheets, loops and coils
- 3' structure of proteins is the combination of 2' structures such that the overall protein structure is formed.
- 4' protein structure is formed when two or more proteins complex together.
- X-Ray Crystallography and NMR Spectroscopy are used to find the structures of proteins
- However, these methods are difficult and expensive
- Solution: Prediction of structures

Introduction

- Protein sequence gives rise to its structure
- If another protein which has a similar sequence also has its structure known, the structure of an unknown protein can be predicted based on that similar protein
- So, it is then possible to identify unknown protein structures by just examining the homologous protein sequences

Conclusions:

- Sequence Identity
- Alignment Length

Which combination of identity and alignment length is suitable for best for structure prediction?

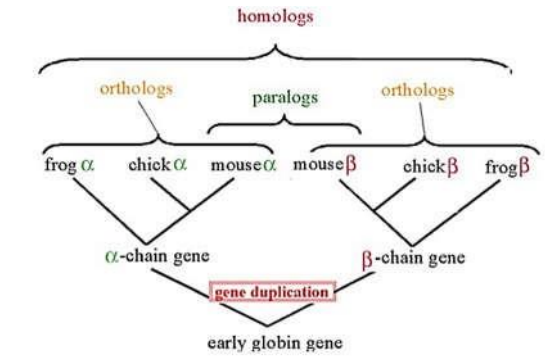
Lecture 218

Homology, Paralogy and Orthology

Background

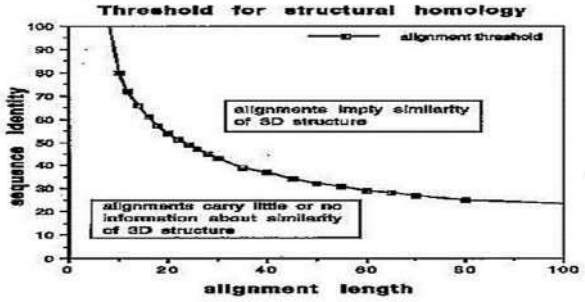
- In homology modelling, proteins with similar 1' sequences are considered
- Given that one of them has its 3' structure known, then the 3' structure of other protein can be predicted

Homology: Paralogy vs. Orthology



http://bioweb.uwlax.edu/GenWeb/Molecular/Bioinformatics/Unit_4/Lab_4-2/lab_4-2.htm

How much homology is required or better?



Conclusions

- Good sequence alignment and identity ensures that homology modelling will give accurate results
- Next, what is the workflow for homology modelling?

Lecture 219

Workflow of Structural Modelling

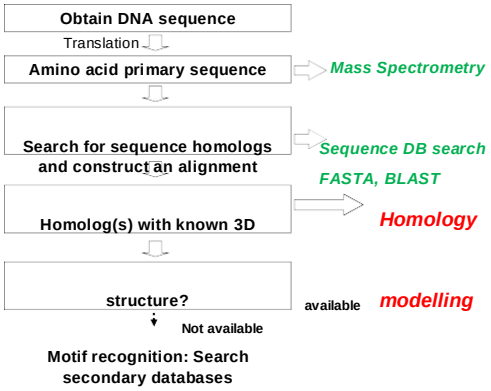
Background

- Homology modelling is used to predict structures of proteins having high sequence similarity with other proteins with known structures!
- Let's consider the workflow of homology modelling

Introduction

Overall, there are three different strategies for structure prediction

1. Homology Modelling
2. Threading/Fold Recognition
3. Ab Initio Modelling



Conclusions

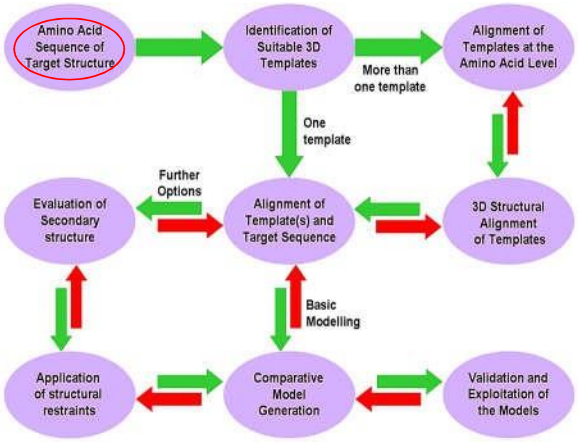
- Next, we will proceed to perform homology modelling
- For that there is a seven step procedure which we will see in the next module.

Homology modeling of the target structure can be done as follows:

1. Template recognition and initial alignment
2. Alignment correction
3. Backbone generation
4. Loop modeling
5. Side-chain modeling
6. Model optimization
7. Model validation

Conclusions

- Homology modelling works in seven steps
- It is a repetitive process
- Next, we will look at each step in detail!



Lecture 220

Seven Steps to Homology Modelling – I

Background

Protein structure can be predicted by 3 methods:

1. Homology Modelling
2. Fold Recognition / Threading
3. Ab Initio Modelling

Introduction

- Let's start by looking at Homology Modelling
- There are seven salient steps in any Homology Modelling pipeline
- Definition of Template (known) & Target (unknown)

Lecture 221

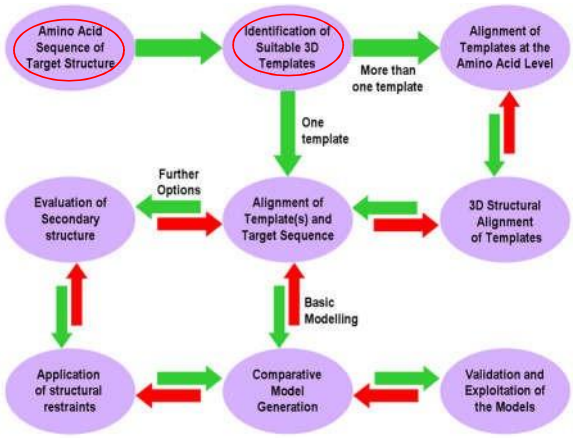
Seven Steps to Homology Modelling – II

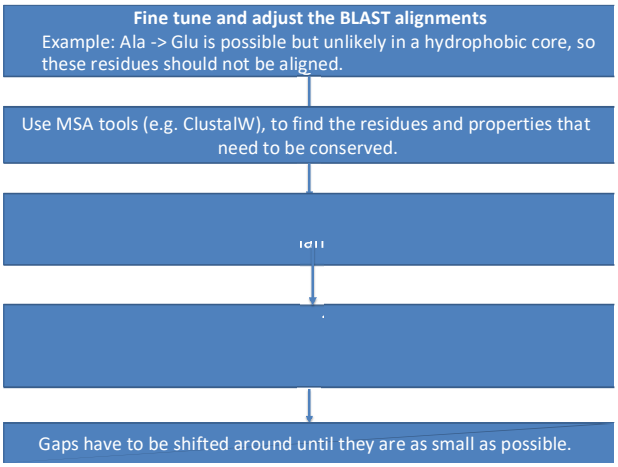
Background

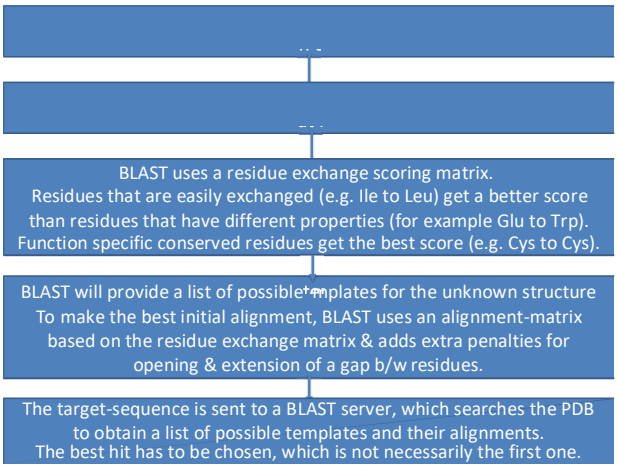
- Template recognition and initial alignment
- Alignment correction
- Backbone generation
- Loop modeling
- Side-chain modeling
- Model optimization
- Model validation

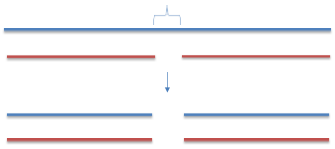
Conclusions

- Now the template and target are selected
- Next, we perform fine-tuning of alignment & introduce corrections to ready the mismatches & gaps.



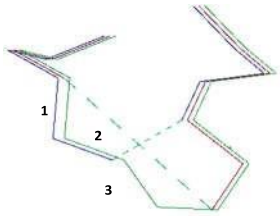






Note: After a deletion of 3 residues a big gap occurs in the RED structure, which was the best alignment.

Remember that deletion of 3 residues left a big gap in the RED structure, which was the best alignment.
After shifting several residues, the gap is much smaller (blue structure) and more likely to be correct.



Lecture 222

Seven Steps to Homology Modelling – III

Background

- Template recognition and initial alignment
- Alignment correction
- Backbone generation

- Loop modeling
- Side-chain modeling
- Model optimization
- Model validation

Conclusions

- The alignment now stands fine tuned and corrected
- Gaps & mismatches have been evaluated & adjusted
- Next step, using this alignment, assemble the backbone

Lecture 223

Seven Steps to Homology Modelling – IV

Background

- Template recognition and initial alignment
- Alignment correction
- Backbone generation

- Loop modeling
- Side-chain modeling
- Model optimization
- Model validation

Next….

- Template recognition and initial alignment
- Alignment correction
- Backbone generation
- Loop modeling
- Side-chain modeling
- Model optimization
- Model validation

The coordinates of the template-backbone are copied to the target.

When the residues are identical, the side-chain coordinates are also copied.

Note that PDB-file may contain small offsets / errors, so try to use multiple similar templates.

When the target (unknown structure) sequence contains a gap, one option is to delete the corresponding residues in template.

- But this creates a fracture in the template!

When the template (known structure) sequence contains a gap, there are no backbone coordinates known for these residues in model. The target backbone has to be cut to insert newer residues.

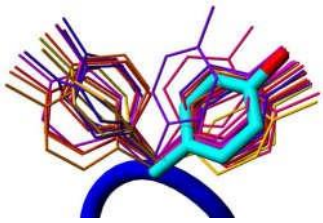
These major changes cannot be modeled in secondary structure elements Hence, place them in loops and strands. Therefore, surface loops are flexible and difficult to predict.

Note that the conserved residues were already copied!
Now, we just need to place the side chains

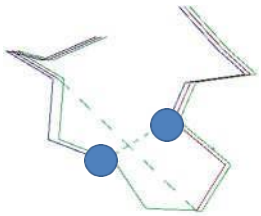
Copy the torsion angles C-alpha/beta to the target!
Rotamers tend to be conserved in homologous proteins and can be predicted as backbone configurations strongly prefer a specific rotamer.

Moreover, libraries of flanking residues can also help estimate the side chain positioning.

The backbone of tyrosine strongly prefers two rotamers & the real side-chain may fit one of them!



One way to handle loops is to take some residues before and after the insertion as "anchor" residues and search the PDB for loops with the same anchor - residues.
The best loop is simply copied in the model.



Conclusions:

- The protein backbone is ready!
- Next, loops were modelled and used to bridge gaps
- Next step, using this backbone & loop choices, place the side-chains.

What is the need for further optimization?
Because the updated side-chains can effect the backbone, and this can effect the structure prediction.

Optimization can be done by performing refinements using Molecular Dynamics simulations of the model.

The model is placed in a force-field and the movements of the molecules are followed in time, this mimics the folding of the protein.

; w

The calculated energy should be as low as possible.

Lecture 224

Seven Steps to Homology Modelling – V

Background

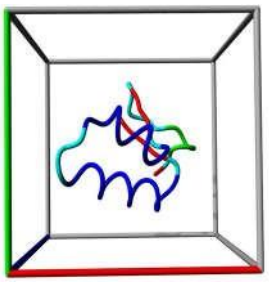
- Template recognition and initial alignment
- Alignment correction

- Backbone generation
- Loop modeling
- Side-chain modeling
- Model optimization
- Model validation

Next….

- Template recognition and initial alignment
- Alignment correction
- Backbone generation
- Loop modeling
- Side-chain modeling
- Model optimization
- Model validation

MD Example Sim: Crambin
(Ethiopian Cabbage Protein)



Conclusions

- Now we have minimized large errors
- However, smaller errors may still exist
- Next step, validate the model that we have constructed!

Lecture 225

Seven Steps to Homology Modelling – VI

Background

- Template recognition and initial alignment
- Alignment correction
- Backbone generation
- Loop modeling
- Side-chain modeling
- Model optimization
- Model validation

Limitations of Homology Modelling

Large Bias towards structure of template
Cannot study conformational changes
Cannot elicit new catalytic/binding sites

```
from modeller import *
from modeller.automodel
  import *
log.verbose()
env = environ()
env.io.atom_files_direct
  ory = './'
a = automodel(
  env,
  alnfile = 'herg.ali',
  knowns = '1q5o',
  sequence = 'herg')
a.starting_model= 1
a.ending_model = 1
a.make()
```

Input Python Script (*.py)

```
>P1;1q5o
structureX: 1q5o : 443 : A : 644 : A ::::
DSSRRQYQEKYKQVEQYMSFHKLPADFRQKIHDYYEHRYQ-
   GKMFDEDSILGELNGPLRE
EIVNFNCRKLVASMPLFANADPNFVTAMLTKLKFEVFQPGD
   YIIREGTIGKKMYFIQHGV
VSVLTKGNKEMKLSDGSYFGEICLL--
   TRGRRTASVRADTYCRLYSLSVDNFNEVLEEYP
MMRRAFETVAIDRLDRIGKKNSIL.*

>P1;herg
sequence: herg : 1 :::::::
YSGTARYHTQMLRVREFIRFHQIPNPLRQRLEEYFQHAWSY
   TNGIDMNAVLKGFPECLQA
DICLHLNRSLLQHCKPFRGATKGCLRALAMKFKTTHAPPGD
   TLVHAGDLLTALYFISRGS
IEILRGDVVVAILGKNDIFGEPLNLYARPGKSNGDVRALTY
   CDLHKIHRDDLLEVLDMYP
EFSDHFWSSLEITFNLRDTN-MIP.*
```

```
ATOM  1  N   ASP A 443   -15.943  41.425  44.702  1.00 44.68
ATOM  2  CA  ASP A 443   -15.424  42.618  45.447  1.00 43.15
ATOM  3  C   ASP A 443   -14.310  43.306  44.686  1.00 41.81
ATOM  4  O   ASP A 443   -14.298  44.528  44.539  1.00 42.61
etc...
```

Template Structure (*.pdb) Sequence Alignment (*.ali)

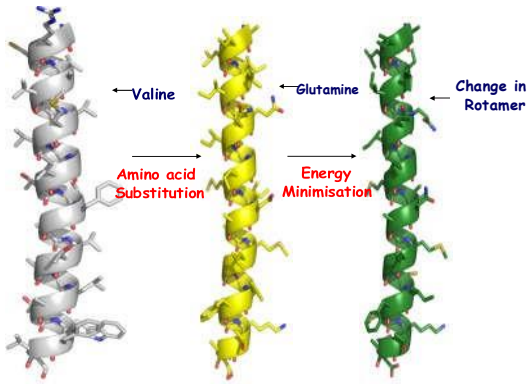
The model with the lowest energy might still be folded completely wrong.

So, the model should be checked again for normal ranges of:
Bumps, Bond angles, Torsion angles, Bond lengths
Other properties, like the distribution of polar/apolar residues, can be compared with real structures.

tolerable. But when an error occurs in the active site, one should reconsider the template and/or alignment.

Conclusions

- So how can we overcome such limitations?
- Other strategies include: Threading, and Ab Initio Modelling
- We will also examine online tools for each.



Lecture 226

MODELLER for Homology Modelling

Background

- Template recognition and initial alignment
- Alignment correction
- Backbone generation
- Loop modeling
- Side-chain modeling
- Model optimization
- Model validation

Background

- Modeller is a software for homology modelling
- salilab.org/modeller
- Inputs: Python script file, Sequence alignment & Template (PDB)

- **.log** : log output from the run.
- **.B*** : model generated in the PDB format.
- **.D*** : progress of optimisation.
- **.V*** : violation profile.
- **.ini** : initial model that is generated.
- **.rsr** : restraints in user format.
- **.sch** : schedule file for the optimisation process.

Automated Modelling Servers

Swiss Model: http://swissmodel.expasy.org//SWISS-MODEL.html

Robetta: http://robetta.bakerlab.org/

3D Jigsaw: http://www.bmm.icnet.uk/servers/3djigsaw/

Phyre: http://www.sbg.bio.ic.ac.uk/phyre/

Conclusions:

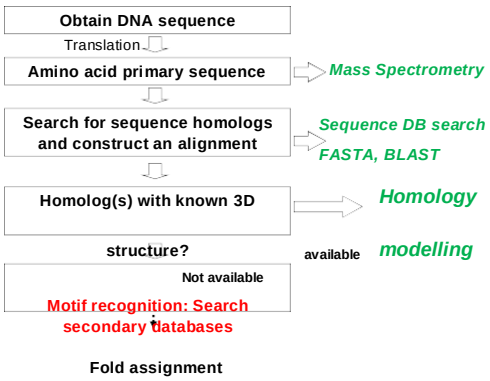
- Homology modelling helps predict protein structures by using prior structural information
- Several tools are available to perform homology modelling in a programmatic or automated way!

Lecture 227

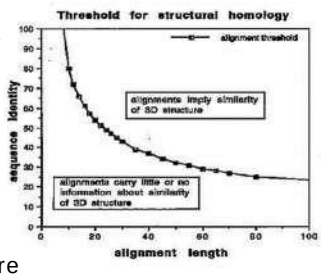
Fold Recognition – Threading I

Background

- Template recognition and initial alignment
- Alignment correction
- Backbone generation
- Loop modeling
- Side-chain modeling
- Model optimization
- Model validation



- A protein fold is defined by the way the secondary structure elements of the structure are arranged relative to each other in space.
- Common folds include 4-helix bundle & the TIM barrel.
- 5,000 stable folds in nature



- Fold recognition: Finding the best fit of a sequence to a set of candidate folds.

Conclusions: Fold recognition or Threading is a technique for predicting protein structures.

- It is useful in cases where homology modelling fails to predict quality structures.

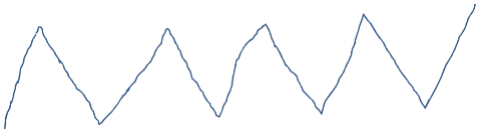
Lecture 228

Fold Recognition – Threading II

Background

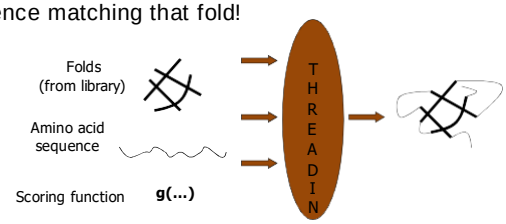
- Fold recognition is also called Threading
- Technique for predicting protein structures
- Employed when homology modelling cannot predict quality structures

Find the best way to "mount" the residue sequence of one protein onto a known protein structure!



The process of threading

- In the process of "Threading", we mount an amino acid sequence on to the backbone of template structures in a folds library
- Each step is "drag" along the sequence (MQVKLFTY...) through each location of each template fold.
- Then, for each fold, we must compute the fitness of sequence matching that fold!



Inputs and outputs of threading G

Conclusions:

- Threading involves "passing" the amino acid sequence through each fold in the database
- The best match is computed using a scoring function.

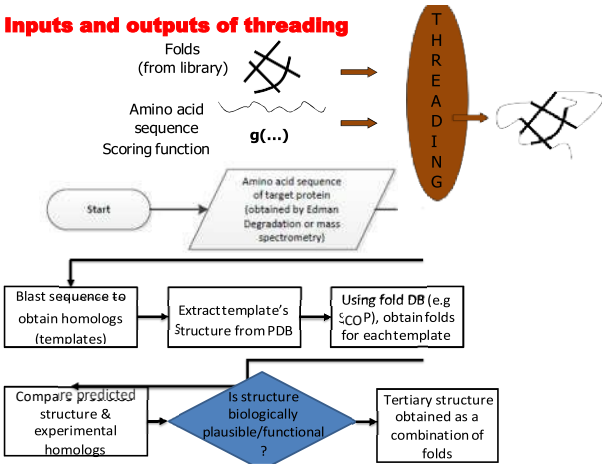
Lecture 229

Fold Recognition – Threading III

- Threading involves "passing" the amino acid sequence through each fold in the database
- The best match is computed using a scoring function
- Flowchart

Conclusions

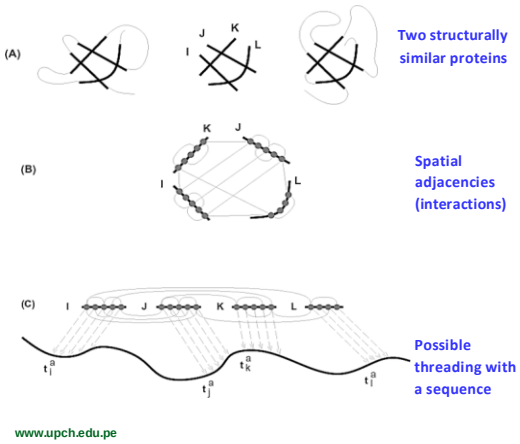
- Combinations of secondary structures come together to form the best prediction
- Scoring typically involves using a Z-Score function based on energy of a molecule



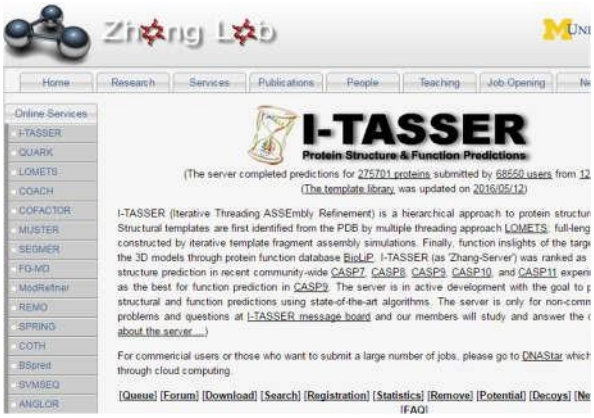
Iterative threading assembly refinement (I-TASSER) server: Software for automated protein structure & function prediction based on the sequence-to-structure-to-function. Steps:

- Starts from amino acid sequence
- i-TASSER first generates 3D atomic models from multiple threading alignments & iterative structural assembly simulations.
- The function of the protein is then inferred by structurally matching the 3D models with other known proteins.
- Outputs full-length secondary & tertiary structures & functional annotations on ligand-binding sites

An estimate of accuracy of the predictions is provided based on the confidence score of the modeling.



Link: http://zhanglab.ccmb.med.umich.edu/I-TASSER/



Lecture 230

Online Tools for Threading - iTasser

Background:

- Threading involves "passing" the amino acid sequence through each fold in the database
- The best match is computed using a scoring function
- iTASSER

Conclusions:

- iTASSER helps thread amino acid sequences on fold and secondary structure databases
- It also helps predict function of structures output.



Lecture 231

Advantages & Disadvantages of Threading

- Fold recognition or Threading is a technique for predicting protein structures.
- It is useful in cases where homology modelling fails to predict quality structures.

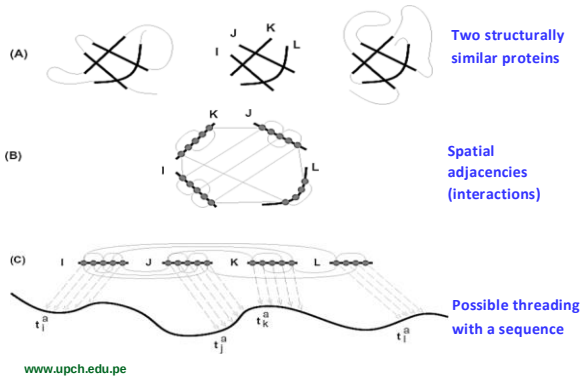
Advantages:

- Threading helps predict secondary structures of proteins towards tertiary structure prediction
- For the "Twilight Zone" with low alignment quality and identity, threading is useful

Disadvantages:

- Novel proteins cannot be predicted using threading
- Fewer than 30% of the predicted first hits are true remote homologues
- Validation of each result is necessary

Inputs and outputs of threading



Lecture 232

3D-1D Bowie Algorithm

Background

- Homology employed high alignment scores
- Threading worked by creating combinations of primary sequences and corresponding secondary structures

Introduction

- Proposed by Bowie et al in 1991
- Converts 3D structure into a 1-D string profile for each structure in the fold library
- Align the target sequence to these profiles

Inputs and outputs of 3D-1D

- Identify amino acids based on: protein core, side chain positioning, solubility etc. (6 in all)
- Part of secondary structure including α-helix, β-sheet etc (3 in all)
- Total of 3 x 6 = 18 distinct states

P = prob. of finding amino acid (a) in environment (j)

$P_{a:j}$ = probability of finding (a) anywhere

Maximize sum of scores for the fold:

$$s = \sum_{aj} \log\left(\frac{P_{a:j}}{P_a}\right)$$

Conclusion

- 3D-1D methods convert structure and environment information into "profiles"
- Score for each amino acid is computed for each profile

Lecture 233

Introduction to Ab Initio Modelling

Background

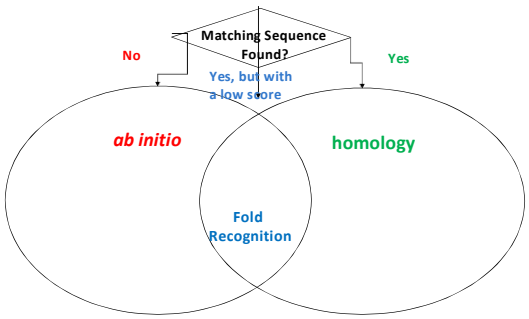
- Ab initio methods have Anfinsen'sthermodynamic hypothesis at the center
- These methods attempt to identify the structure with minimum free energy

Need for Ab Initio Modelling

- Applicable to any sequence
- Not very accurate biologically
- Accuracy & applicability are limited by our understanding of the protein folding problem

Limitation

- Computationally expensive
- Suitable for proteins with less than 100 residues



Conclusion

- Ab initio methods rely on computing the energies of folded proteins
- The protein structures with the lowest energy are deemed as plausible predictions

Lecture 234

Rationale of Ab Initio Modelling

- Ab initio methods rely on computing the energies of folded proteins
- The protein structures with the lowest energy are declared as plausible predictions

Rationale

- Sometimes it so happens that even slightly homologous proteins may not be available.
- This renders homology modelling and threading/fold recognition as futile.
- Also, newer protein structures continue to be discovered every day
- These could not have been identified by methods which only rely on matching with available structures

Simulation of the Folding Process

1. Build an accurate initial model (including energy & forces).
2. Accurately simulate the dynamics of the protein folding process.
3. The native structure will steadily emerge.

Conclusion:

- Start with an energy function
- Fold structures in order to obtain the most stable structure
- This structure will have the minimum energy

Lecture 235

Strategies for Ab Initio Modelling

Background

- Ab initio methods base their predictions on physical models of folding mechanisms
- Stabilization is measured by energy released during the folding process.

Energy Optimization in Ab Initio Modelling

1. Start with a rough initial model.
2. Define an energy function mapping structures to energy values. We have to minimize this later!!
3. Solve the computational problem of finding the global minimum.

- Lastly, homology / fold recognition predict protein structures without computing fundamental physical/chemical properties of the mechanisms and driving forces in structure formation

Conclusion

- Ab initio methods, in contrast, base their predictions on physical models for these mechanisms
- Energy released during the folding process is computed for predicting structure.

Lecture 236

Energy States of Folded Proteins

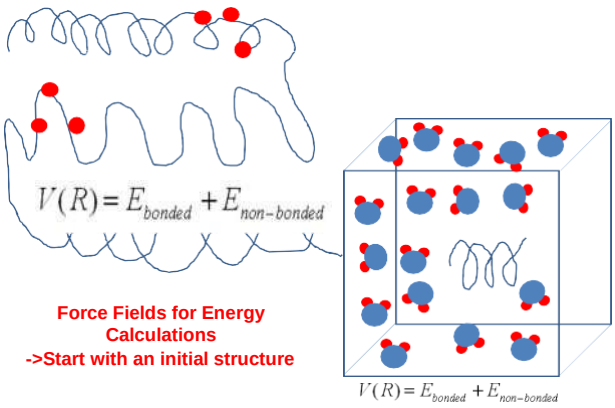
Background

- Ab initio methods predict protein structures by folding proteins based on each constituent atom's volume, charge, mass etc.

Conclusion

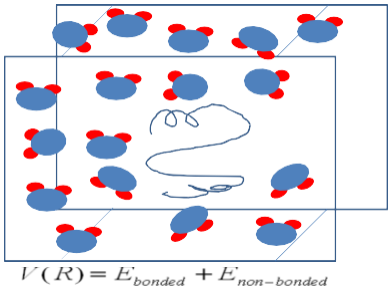
- The protein structure reporting lowest energy is selected to be the optimal structure
- How easy is it to compute the "really" lowest energy of a folded protein?

Energies of Bonded Atoms vs. Nonbonded Atoms

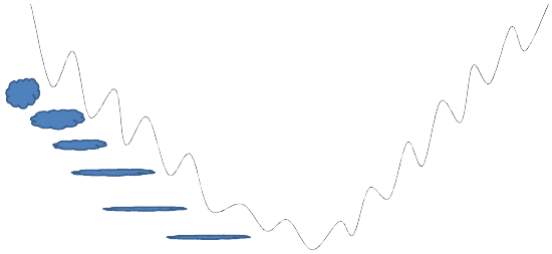


Force Fields for Energy Calculations
->Start with an initial structure

Force Fields for Energy Calculations



Global Energy Optimization helps find global minimum



Lecture 237

Local versus Global Minima

- The protein structure reporting lowest energy is selected to be the optimal structure
- How easy is it to compute the "really" lowest energy of a folded protein?

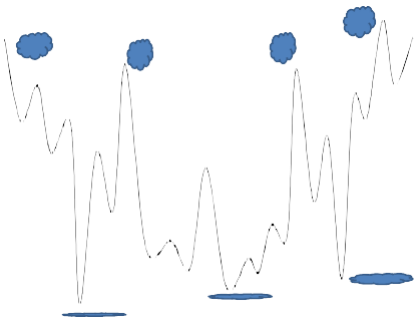
Best Case Energy Function

- Clear energy minimum in the native structure
- Viable path towards this minimum
- Global optimization finds the most stable structure

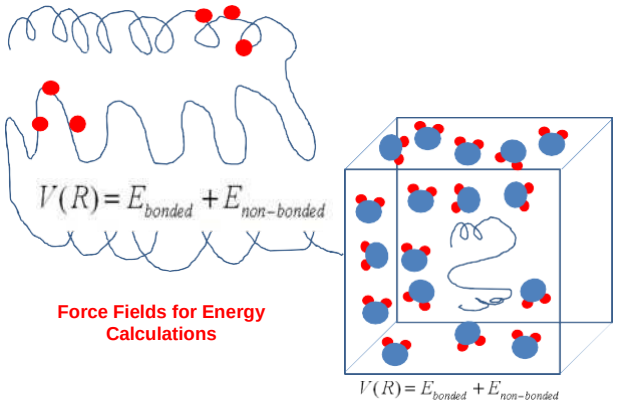
Optimal Energy Function

- Easier to design and compute
- Native structure not always at the global minimum
- No clear way of choosing among alternative structures that are generated

Global Energy Optimization helps find global minimum



Energies of Bonded Atoms vs. Nonbonded Atoms



Force Fields for Energy Calculations

Lecture 238

Pros and Cons of Ab Initio Modelling

Background

- Native structure not always at the global minimum
- No clear way of choosing among alternative structures that are generated

Advantages

- Ab Initio methods can fold any target sequence using only physical atomic properties
- Predictions are mostly accurate and correctly describe the natural folding process.

Disadvantages

- Ab initio methods are the very difficult to design (energy function)
- These methods are slow due to the huge possibilities

Disadvantages

- An order of 10^{12} steps are needed to simulate protein folding for medium sized protein structures

Challenges in Ab Initio Modelling:

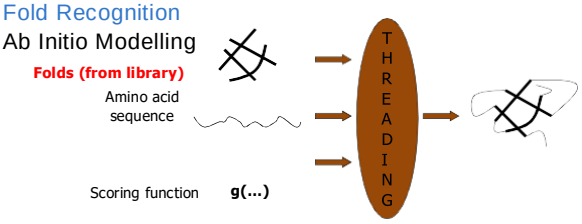
- Very hard to accurately describe energy functions that can reliably discriminate native and non-native structures.
- Enormous amount of computations.

Lecture 240

Summary of Structural Modelling – II

Strategies for Structural Modelling

- Homology Modelling
- Fold Recognition
- Ab Initio Modelling



Conclusion

- For low identity and alignment scores, a "Twilight zone" for structure prediction exists
- Fold recognition / threading is useful in such cases

Lecture 239

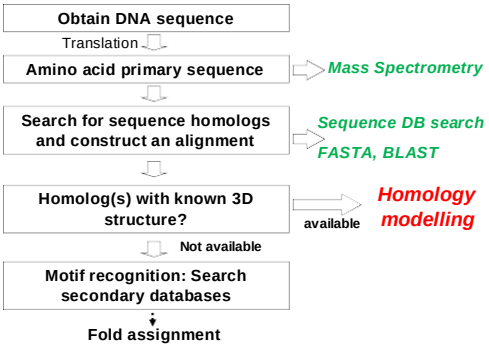
Summary of Structural Modelling – I

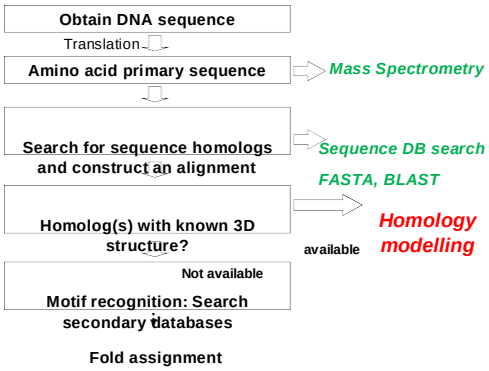
Strategies for Structural Modelling

- Homology Modelling
- Fold Recognition
- Ab Initio Modelling

Homology modeling of the target structure can be done as follows:

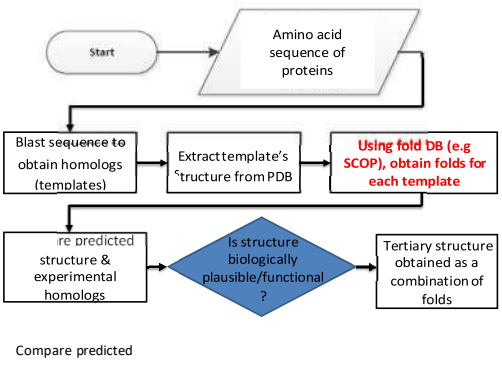
1. Template recognition and initial alignment
2. Alignment correction
3. Backbone generation
4. Loop modeling
5. Side-chain modeling
6. Model optimization
7. Model validation





Conclusion

- Homology modelling is performed in cases of high identity and alignment score
- For the "Twilight zone", other strategies are employed



Lecture 241

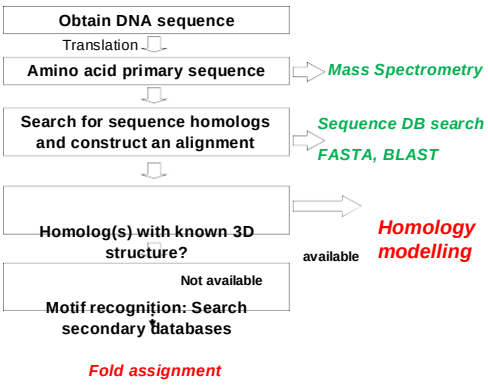
Summary of Structural Modelling – III

Strategies for Structural Modelling

- Homology Modelling
- Fold Recognition
- Ab Initio Modelling

Energy Optimization in Ab Initio Modelling

1. Start with a rough initial model.
2. Define an energy function mapping structures to energy values. We have to minimize this later!!
3. Solve the computational problem of finding the global minimum.



Simulation of the Folding Process

1. Build an accurate initial model (including energy and forces).
2. Accurately simulate the dynamics of the protein folding process.
3. The native structure will steadily emerge.

Conclusion

- For cases where even the fold libraries do not give any high scoring matches, Ab Initio strategies can help model the structure.
- However, this is a complex and computationally expensive process.

Lecture 242

Review of Sequence Analysis

Important Concepts

How do we sequence:

- Genomes
- Proteomes

How do we compare sequences:

- Pair-wise Sequence Alignment
- Multiple Sequence Alignment

Types of Alignments:

Global Alignment (Needle Wunsch)
Local Alignment (Smith Waterman)

Advanced Tools:

Fast Alignment (FASTA)
Basic Local Alignment Search Tool (BLAST)

Important Concepts

Databases:

GenBank
UniProt

Important Concepts

Online Portals:

Ensemble
Expasy
UniProtKB

Lecture 243

Review of Phylogenetics

Important Concepts

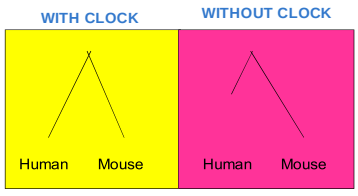
- Molecular Evolution
- Insertions
- Deletions
- Substitutions

Important Concepts

- Phylogenetic Trees
- Scaled Trees
- Unscaled Trees
- Phylogenetic Trees
- Rooted Trees
- Unrooted Trees

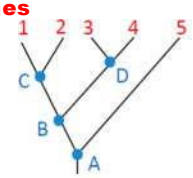
Edge length with and without a clock!

Since the rate of evolution in species is different, it needs to be considered!



UPGMA: Unweighted Pair – Group Method using arithmetic Averages

- Two sequences with with the **shortest evolutionary distance** between them are considered
- These sequences will be **the last to diverge**, and **represented by the most recent internal node.**



Important Concepts

Clustering Vs. Non-clustering Methods:
UPGMA is a clustering method
Maximum Parsimony etc are non-clustering methods (not included in this course).

Lecture 244

Review of Protein Sequencing

Important Concepts

- Techniques of protein sequencing
- Edman Degradation
- Mass Spectrometry
- Protein Ionization
- Mass Analysis
- Protein Fragmentation

Important Concepts

- MS1
- MS2

Important Concepts

- Estimating and scoring whole protein mass
- Extracting & Scoring Peptide Sequence Tags
- Searching Post-translational Modifications

Important Concepts

Composite Scoring Schemes
Online tools:

- Mascot
- Sequest
- Prosight PC

Lecture 245

Review of RNA Structure Prediction

Important Concepts

- Role of RNA in biological processes
- Atomic Force Microscopy
- Need for Structure Prediction

RNA Secondary Structures

- Hairpin Loops
- Bulges
- Helices
- Intersection
- Conceptual basis for structure prediction
- Energy is released as a result of nucleotide coupling
- Folded RNA is more stable in terms of energy
- Algorithms for predicting RNA Structure
- Dot plot
- Zuker's Algorithm
- Martinez Algorithm
- Nussinov Jacob Algorithm
- RNA Structure Databases
- Online tools for predicting structures given a sequence

Lecture 246

Review of Protein Structures

Important Concepts

Protein Structures are generally of four types:

- Primary
- Secondary
- Tertiary
- Quaternary

Important Concepts

Techniques for determining protein structures

- X-Ray Crystallography
- NMR Spectroscopy
- Why number of known protein sequences is much larger as compared to known proteins structures?

Types of Protein Secondary Structures

- Helices
- Beta Sheets
- Coils
- Loops
- Foundation of structure prediction algorithms
- Propensities of certain amino acids to form specific secondary structures
- Algorithm for predicting protein structures
- Chou Fasman Algorithm
- Protein Structure Database - PDB
- Online tools for predicting structures by using proteins sequences.

Lecture 247

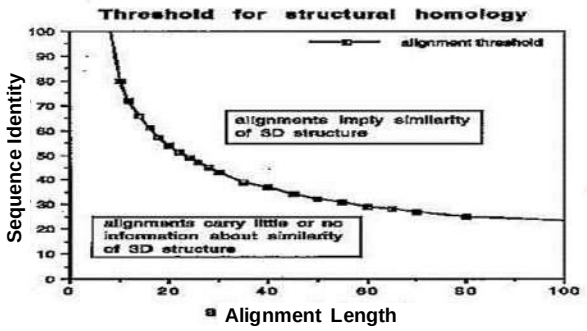
Review of Homology Modelling

Important Concepts

Four Strata of Protein Structures

- Primary
- Secondary
- Tertiary
- Quaternary
- Justification for homology modelling
- Number of known protein sequences is much larger as compared to known proteins structures

Three Strategies for Structure Prediction
- Homology Modelling
- Fold Recognition
- Ab Initio Modelling



Lecture 249

Advanced Follow-up Courses

Important Concepts
- We looked into the foundations of Bioinformatics
- However, each topic that was studied has a undergone a lot of development.
- For advanced study in Genomics, you may take "Computational Genomics" course
- Topics: Genome Assembly, Gene Finding, Annotation, GWAS etc
- For advanced study in Proteomics, you may take "Computational Proteomics" course.
- Topics: Protein Sequencing, PTM search, Structure Modelling and PPI studies

Important Concepts
- Protein Structure Database - PDB
- Online tools for predicting structures such as MODELLER and iTASSER

Lecture 248

Conclusions from this Course

Important Concepts
- Definition of Bioinformatics
- Need for Bioinformatics
- Areas within Bioinformatics
- Bioinformatics as an interdisciplinary area
- Need to store, process and analyze biological data
- Requirement of newer faster algorithms

- For advanced study in Integrative Biology, you may take "Systems Biology" course.
- Topics: Metabolomics, Transcriptomics, Network Biology etc

Also, now there are cutting edge courses on:
- Nano-Bio-IT
- Computational Drug Design
- Personalized Medicine

Lecture 250

Careers in Bioinformatics

Background
- Pakistan as an infrastructure-limited country
- The onset of digital revolution
- Emergence of data as the most precious commodity, globally

Specific areas focused were:
- Comparing sequences
- Comparing structures
- Predicting structures

We looked at:
- Algorithms
- Databases
- Online Tools for each topic.

Important Concepts
- We studied the basic algorithms for each topic
- With evolution and growth of Bioinformatics, newer and better algorithms are now also available!

- Specifically, health data as a key commodity of the future
- Health and disease as the primordial challenge of mankind
- Unique opportunity for us in Pakistan
- Bioinformatics requires two things

1. Smart mind
2. Internet connected computer

One man company
- You can take public databases and design drugs!
- One man vs. Roche?

BIGDATA
- You can make a startup company which manages and process health BIGDATA!
- All it needs is basic software development skills coupled with Bioinformatics.

The next disruption

- The next Google, Facebook and Uber is going to emerge from Health and Bioinformatics
- Pharmaceutical companies are investing into bioinformatics human resource development

Jobs Market

- Pharmaceutical Giants
- Research Centers & Universities
- Hospital & Diagnostic IT departments
- Your own startup company